

Determination of research reactor fuel burnup



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FOREWORD

This report was prepared by a Consultants Group which met during 12-15 June 1989 at the Jozef Stefan Institute, Yugoslavia, and during 11-13 July 1990 at the IAEA Headquarters in Vienna, Austria, with subsequent contributions from the Consultants. The Physics Section of the Department of Research and Isotopes was responsible for the co-ordination and compilation of the report.

This report is intended to provide information to research reactor operators and managers on the different, most commonly used methods of determining research reactor fuel burnup: 1) reactor physics calculations, 2) measurement of reactivity effects, and 3) gamma ray spectrometry. The advantages and disadvantages of each method are discussed. References are provided to assist the reactor operator planning to establish a programme for burnup determination of fuel. Destructive techniques are not included since such techniques are expensive, time consuming, and not normally performed by the reactor operators.

The availability of analysed burnup data is an essential first step in any systematic approach to enhancement of economics, safety and performance of a research reactor. The motivation to undertake such studies are:

- Increased cost of reactor fuels
- Increased emphasis on reactor safety
- Demand for higher neutron fluxes.

The necessity for fuel and core management for more efficient fuel utilization by increasing burnup consistent with considerations of safety and performance is evident. To meet the demand for higher fluxes, using LEU fuels, it is essential that detailed analyses of the new core be performed to optimize the available neutron fluxes. Confidence in core management schemes based on intuition becomes much lower and less reliable, particularly for reactors using mixed cores of HEU and LEU fuels. Burnup determinations provide essential input to a systematic approach to the solution of the problem.

EDITORIAL NOTE

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1. INTRODUCTION

The availability of burnup data is an essential first step in any systematic approach to the enhancement of safety, economics, and performance of a research reactor. The motivation to undertake such studies are:

- Increased emphasis on reactor safety
- Increased cost of reactor fuels
- Demand for higher neutron fluxes.

As a consequence of the international effort to reduce the traffic of high enriched uranium (HEU) fuels, low enriched uranium (LEU) fuels were developed [8]. These new LEU fuels are more expensive than the previously used HEU fuels.

The necessity for fuel and core management for more efficient fuel utilization by increasing burnup consistent with consideration of safety and performance became evident. To meet the demand for higher fluxes, using LEU fuels, it became essential that detailed analyses of the new core be performed to optimize the available neutron flux densities [8], [6], [2]. The problem is complicated by the fact that many reactor operators are now using mixed cores of HEU and LEU fuel elements making the analysis of these cores more complex. Confidence in core management schemes based on hand calculations and experience becomes much lower and less reliable and can be in some cases dangerous [3]. Burnup determinations provide essential input to any systematic approach to the solution of the problem.

This report is intended to provide information to research reactor operators and managers on the different, most commonly used methods of determining research reactor fuel burnup: 1) reactor physics calculations, 2) measurement of reactivity effects, and 3) gamma ray spectrometry. The advantages and disadvantages of each method are discussed. References are provided to assist the reactor operator planning to establish a programme for determination of fuel burnup. Destructive techniques are not included since such techniques are expensive, time consuming, and not normally performed by the reactor operators.

Three separate and distinct methods for making non-destructive determination of the burnup of a fuel element are presented in this report. In both reactivity effects and gamma ray spectrometry cases, the method depends upon the calculations which support and supplement the measurements. The gamma ray spectrometry method, which measures specific fission products, is the most direct way for measuring burnup. This is true because the corresponding calculations that are required (fission product inventory as a function of burnup), are less subject to errors than calculations supporting the other two methods.

The reactivity measurement method follows the normal type of calculations and measurements that should be performed at a research reactor for both fuel management and experiment design support activities [12]. This method probably involves greater uncertainties, but has the advantage of providing the data needed to refuel the reactor or establish core configurations for specific experiments. It is important to include an unirradiated (fresh) fuel element or, in general, at least one element with known burnup in making these measurements to provide a benchmark data point.

A knowledge of the flux spectrum and power distribution are important elements in the burnup calculation performed for both types of measurements. The more accurate the computer codes used to support the experiment, the more accurate will be the results. It is also essential that the thermal power calibration of the reactor be performed as accurately as possible.

In this report, TRIGA fuel elements are used in most examples to describe the methods [11]. The same techniques however can be used for research reactors which use different types of fuel elements, e.g. plate type fuel elements. Such methods may be more difficult to perform for these research reactors because of the fuel geometry. The gamma scanning method for the plate type will be simpler if each fuel plate can be measured separately. This will require dismantling the fuel assembly. Otherwise tomography may be required. When reactivity measurements are made, the variation in reactivity between fuel elements may be larger and the effect of gradient depletion within the fuel element may introduce larger errors unless appropriately analyzed. Nevertheless, if due attention is paid in making the studies, both methods should provide useful data in determining the burnup of fuel elements for research reactors.

2. REACTOR PHYSICS CALCULATIONS

The determination of the burnup of research reactor fuel elements using reactor physics methods involve both calculational and experimental methods.

Calculational methods are more commonly used in practice than experimental methods, especially when the burnup of a large number of fuel elements in the core has to be determined or routinely updated. Experimental methods are not practical in such cases because they influence reactor operation due to special experimental requirements, i.e. no xenon, constant temperature, etc. Experimental methods are mainly used in support of calculations and only in special cases as the main method, e.g. when the calculations are not available or not reliable. Their main importance is in providing independent reference results for checking and correcting the results of burnup calculations.

The calculations and experiments described here are the types normally performed in support of fuel management and core-experiment design studies [4]. All of the techniques involved are relatively simple and easy to perform; however, in regard to the calculations one must have the proper computer codes which incorporate library cross sections of all important nuclides in the core.

In principle, there is a wide selection of reactor physics codes which could be used for research reactor burnup calculations. However, this report will treat mainly those which are widely available (e.g. through NEA Data Bank) and which have been adapted and practically used for research reactor calculations. Typical representatives of such codes are presented in Table 1. Some of them are briefly described in the following text together with general principles of practical research reactor calculations.

Practical burnup calculations are usually performed in two steps. In the first step, which is commonly known as unit cell calculations, the reactor is virtually divided into a number of geometrically equivalent units called unit cells. The definition of unit cells in a reactor with periodic fuel rod lattice (Fig. 1) is straightforward: a unit cell is identical to the smallest geometrical unit which can be periodically repeated. In a periodic lattice it is a hexagon or a square usually, but not necessarily having a fuel rod in

TABLE 1. TYPICAL COMPUTER CODES USED FOR RESEARCH REACTOR BURNUP CALCULATIONS

Code	Description/Geometry	Computer		Application	Availability
		Mainframe	Small		
UNIT CELL CALCULATIONS					
LEOPARD	1-D unit cell	IBM, CDC, VAX	micro-VAX	MTR, TRIGA	NEA Data Bank
WIMS-D/4	1-D unit cell 2-D multi cell	IBM, CDC, VAX	micro-VAX, IBM-PC(AT-386, AT-286)	TRIGA, MTR	NEA Data Bank/1/,/9/
GLOBAL (DIFFUSION) CALCULATIONS					
EXTERMINATOR-2	multigroup, 2-D finite difference	IBM, CDC, VAX		MTR, TRIGA	NEA Data Bank
SIXTUS	multigroup, 2-D hexagonal geometry	CDC, VAX	micro-VAX	TRIGA	authors /10/
NEXT	multigroup, 3-D, X-Y-Z, nodal	IBM, CDC, VAX	micro-VAX, IBM-PC	MTR, TRIGA	NEA Data Bank /5/ authors
INTEGRATED PACKAGES					
TRIGAP	2-group 1-D radial	IBM, CDC, VAX	micro-VAX, IBM-PC	TRIGA	NEA Data Bank /7/ authors
2DB-LEOPARD	multigroup 2-D, X-Y	IBM, CDC, VAX		MTR	NEA Data Bank

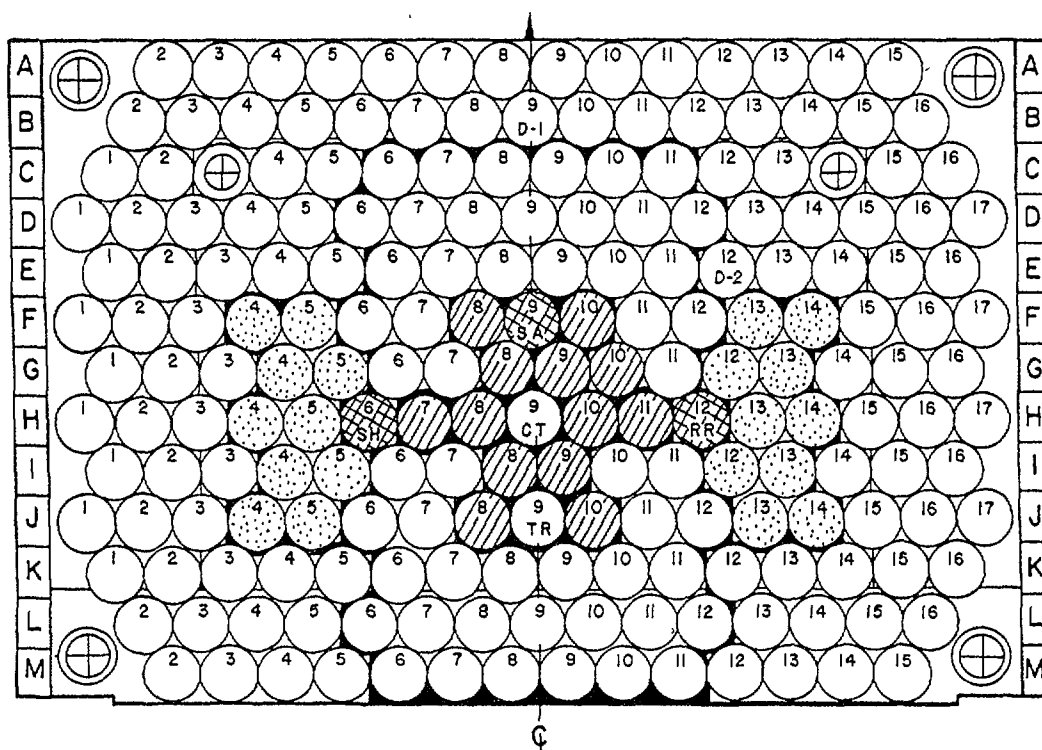


FIG. 1. PSBR core configuration for reactivity measurements — example of hexagonal fuel rod lattice.

the centre: according to the core geometry, certain unit cells contain structural parts of the core other than fuel or they simply contain only water if the fuel rod is not present at that location. In case of non-periodic fuel rod lattice (e.g. cylindrical TRIGA, Fig. 2), the unit cells are more difficult to define. Usually they can be defined so that they correspond to grid locations and have equal volumes.

After the unit cells have been defined, their effective homogenized few-group macroscopic cross sections are calculated by using a unit cell code appropriate for the specific cell geometry and structure. Care must be taken when selecting the code and its data library since the fuel of some research reactors incorporates nuclides not found in standard data libraries that are generated primarily for power reactor calculations. This requirement considerably reduces the choice of unit cell codes appropriate for research reactor calculations (see Table 1). Typical representatives of such codes are LEOPARD and WIMS-D/4 [1].

Unit cell calculations are performed to generate the effective few group homogeneous cross sections for all unit cells in the reactor. These few group cross sections are then used by the global reactor code to calculate power density and fuel burnup distribution in the core as shown in the flow chart on Fig. 3. This is the second important step of the reactor burnup calculation, often called global or diffusion calculation.

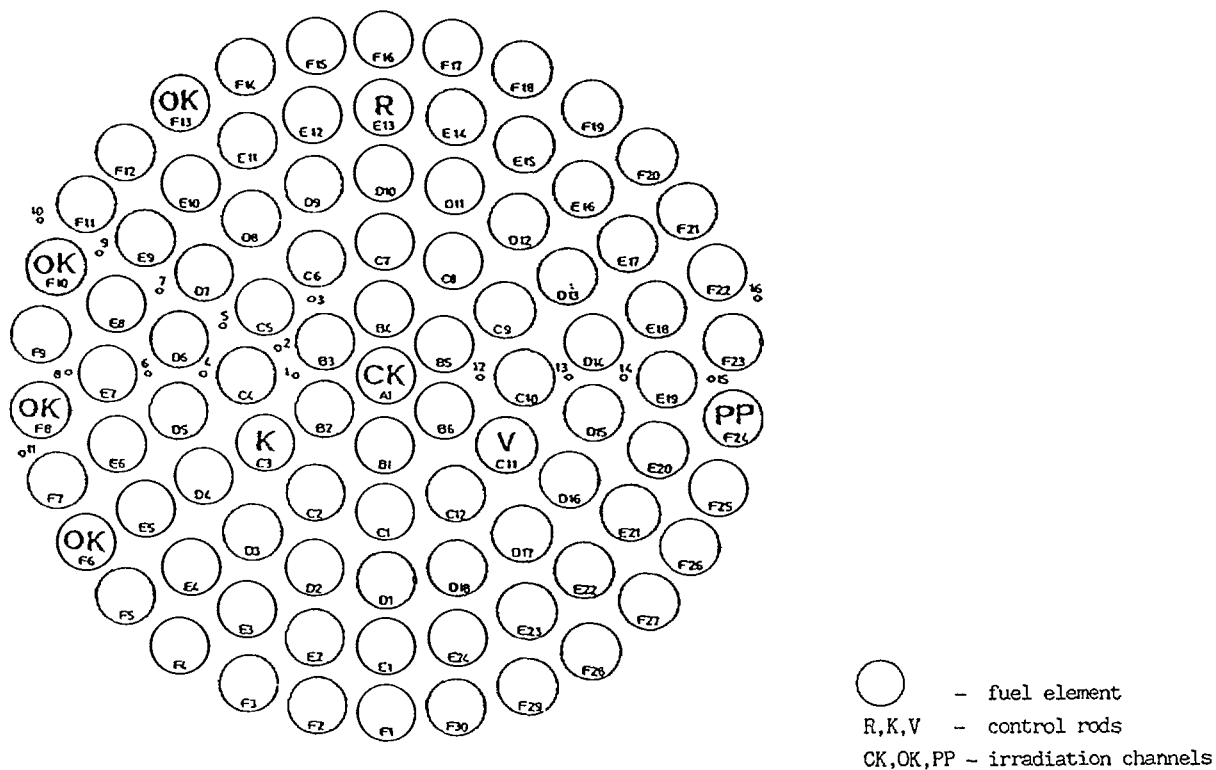


FIG. 2. TRIGA Mark II core configuration — example of non-periodic fuel rod lattice.

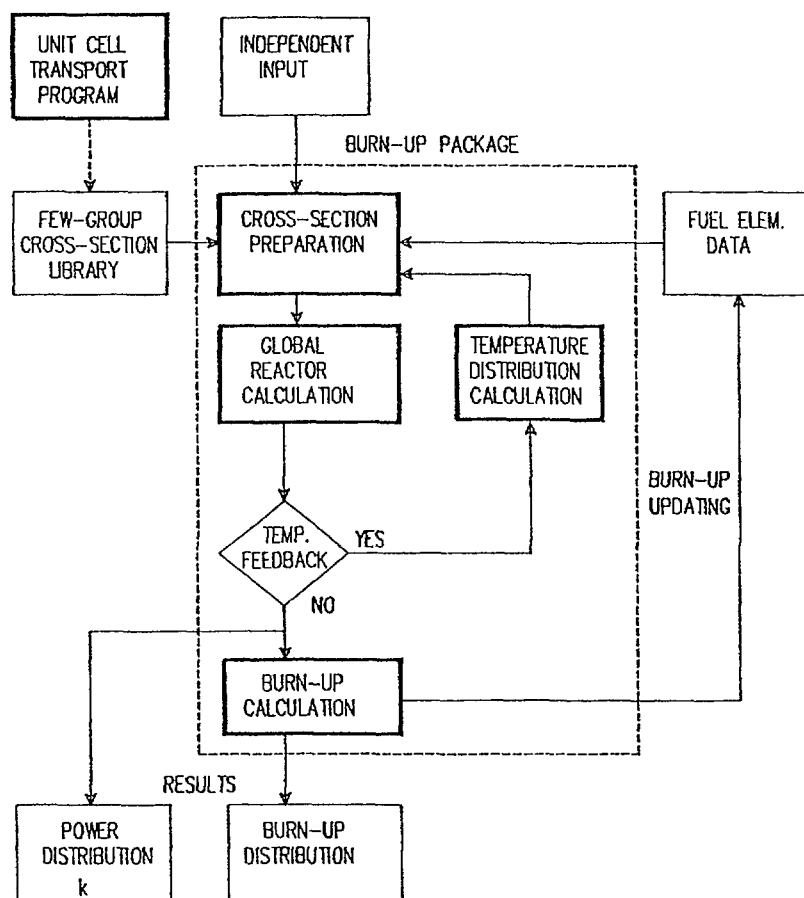


FIG. 3. Schematic diagram of the burnup calculation.

The purpose of the global reactor calculations is to determine the multiplication factor and the detailed power distribution of the core. They are performed in one or two-dimensional diffusion approximation depending on the geometry of the core. Several computer programmes are available for these purposes, based on classical finite differences or modern nodal methods. However, in spite of nodal codes' higher calculational efficiency, the finite differences codes seem to be more appropriate for research reactor calculations, since they offer more flexibility in geometry modelling. A good example of such codes is EXTERMINATOR 2.

The macroscopic cross sections for global reactor calculations are usually provided in the form of data libraries. Data library consists of diffusion coefficients, macroscopic cross sections (absorption, removal, fission), fission energy yields and other data relevant for multigroup diffusion calculation of the whole reactor. According to the diffusion approximation, the cross sections are provided in homogenized form for all regions in the reactor: fuel rods, reflector region, void locations, control rods, irradiation channels, etc. The data are tabulated as functions of important parameters, which change during operation, e.g. burnup and temperature.

The preparation of the few-group cross section library is usually not an integral part of the burnup calculations. The library is prepared separately and in general form, so that it can be used in different burnup calculations.

The computer codes, which are used for few-group effective cross section calculations, are usually much more complicated and bigger than those used for burnup calculations, because they are based on multigroup transport theory. They require a larger computer and deeper knowledge of reactor physics, which makes their use by a typical research reactor staff less probable.

It is convenient that the few-group cross sections, which are required as input for global reactor calculations, be prepared in an automatic way by a special routine. The function of this routine is to interpolate data in the library according to the given burnup, temperature and other parameters, for all unit cells which include fuel elements and other parts of the reactor. The burnup values and other parameters are stored on a special file, which contains complete information about all elements in the core: identification number, type, burnup etc. For burnup calculations of a specified reactor core with defined loading pattern, it is necessary that the fuel element data file contains information only for the elements which are used in the core.

However, it is very practical that it contains also data for all other elements which are momentarily not used in the core but are still interesting for fuel management. In this way, the file becomes a "logbook" of fuel burnup history.

The global reactor calculations are in general repeated several times in the same burnup step, in order to self-adjust power and temperature distributions in the core. For cross section preparation, the temperature of all fuel elements must be known. However, they can be calculated only after the power distribution is given. The problem is solved in an iterative way: a temperature distribution is guessed, cross sections are prepared, power distribution calculated, new temperature distribution is calculated and the whole cycle is repeated until the difference between two consecutive power distributions becomes negligible. In practice at least two iterations are required, especially in mixed core calculations where power distributions are not smooth functions.

The relation between fuel element power and its temperature can be either calculated or taken as a result of measurements. Calculations of this kind are rather complicated and require good thermal-hydraulic codes especially for the reactors which are cooled by natural convection. They are usually not explicitly performed in the chain of reactor burnup calculations. Correlation between power and temperature is instead simply tabulated and built into the cross section preparation codes for typical reactor geometries and operating conditions. The correlation is usually based on results of thermal-hydraulic calculations supported by experimental data. An example of such correlation between fuel element power and average temperature for natural convection TRIGA core is presented in Fig. 4.

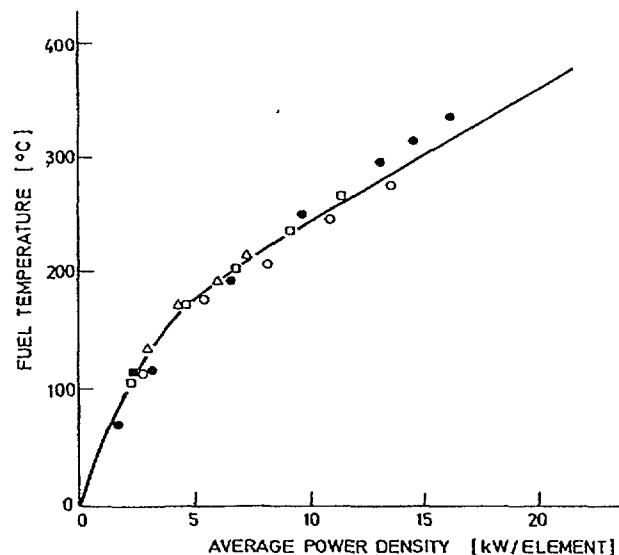


FIG. 4. Fuel temperature (above ambient) as a function of fuel element power for natural convection TRIGA core.

The programmes for cross section global reactor calculations are often integrated in programme packages (circled by a dotted line in Fig.3), in order to optimize data transfer between the codes and to simplify input preparation. Examples of such packages are 2DB-LEOPARD (plate type reactors) and TRIGAP (TRIGA reactors) [7]. They are distributed together with the cross section libraries, which are appropriate for all typical burnup calculations.

The calculational techniques described above are used to determine the burnup in the fuel elements. One of the most important results of the global reactor calculation is the power density distributions, which is usually expressed in terms of power released per fuel element P_j and normalized to the reactor operating power Q :

$$Q = \sum_{j=1}^{N_{FE}} P_j \quad , \quad (1)$$

where N_{FE} is number of fuel elements in the core.

Average power produced per fuel element $\bar{P}$ is defined as

$$\bar{P} = \frac{Q}{N_{FE}} \quad . \quad (2)$$

Eq. (2) is a simple but interesting equation. It shows that even if the reactor power Q remains a constant, the average power per fuel element is inversely related to the number of fuel elements in the core. Thus as fuel elements are added to a core to compensate for the loss of reactivity due to core burnup, the power produced by each element and the average flux density in the core are reduced. Also core lifetime is increased due to lower xenon concentration, but at the expense of a lower neutron flux density in the core.

Let us suppose that the reactor is operating at constant power Q during a time interval Δt . The burnup increment ΔBU_j^E of fuel element j at the end of this interval is

$$\Delta BU_j^E = P_j \Delta t \quad . \quad (3)$$

According to Eq. (3), burnup will be expressed in energy units, e.g. in MWd, if power is measured in MW and time in days. This is denoted by superscript E. The total burnup of a fuel element is calculated by adding the burnup increment to the accumulated burnup in the previous steps. The calculation is started with a fresh fuel element and repeated in steps as long as the element is in the core.

It may be noted that Eq. (3) holds for each element in the core. During the same time step also holds

$$\sum_{j=1}^{N_{FE}} \Delta BU_j^E = \bar{N}_{FE} \cdot P \Delta t = Q \Delta t \quad (4)$$

However, the same equation cannot be used in another step because the number of fuel elements, N_{FE} may be different. For this reason it is inconvenient and not as useful to define quantities like average fuel element burnup or core average burnup. The only integral parameter which can be defined is the accumulated energy produced by the core. All other burnup parameters have their meaning only for individual fuel elements. Burnup is for this reason usually calculated and logged for individual elements. Sometimes groups of fuel elements with the same burnup history are treated together for practical reasons. Such simplifications imply burnup averaging over the group and can be a source of serious errors if not treated carefully.

Another reason why it is impractical to introduce core or region averaged burnup values is the fact, that burnup is commonly expressed in terms of energy produced per mass of heavy isotopes (e.g. MW.d/t) rather than in terms of energy produced per element. Burnup increment of a fuel element j in MW.d/t (denoted by ΔBU_j^M) is obtained, if ΔBU_j^E is divided by the initial mass of uranium M_j , in tonnes,

$$\Delta BU_j^M = \Delta BU_j^E / M_j \quad (5)$$

It is important to note that M_j can be significantly different for different fuel element types in the same core, e.g., fresh standard TRIGA fuel element with 8.5 w/o U concentration contains 190 g of uranium, while fresh LEU TRIGA fuel elements contain 495 g. For illustration let us suppose that standard and LEU fuel elements produce the same power, so that their burnup in energy units is the same. However, we see from Eq. (5) that their burnup in MW.d/t is 2.5 times different.

Several computer codes use MW.d/t burnup units and require so called "specific power" in MW/t as input parameter (e.g. WIMS). Specific power is calculated so that the power of an element P_j is divided by the initial mass of heavy isotopes M_j of the same element. However simple, this definition is a constant source of trivial errors. Another source of errors is that instead of initial mass of heavy nuclides its current value is taken for M_j in burnup calculations. This often happens when the burnup calculations are started with partly depleted fuel.

The above equations show the importance of following each core configuration change to determine the burnup of each fuel element during its history in the core. The most important operating parameters, which are required besides fuel element data for burnup calculations, are core loading pattern, operating power, and time interval of constant power. They are taken from the reactor operation log-book. The operating power may become a source of error if the calorimetric calibration is not adequate. Surprisingly, the loading pattern can also indirectly be a source of error. In practice the loading pattern may be changed rather frequently (10 times per year) making burnup calculations tedious, especially if the burnup history of several years is to be repeated. For practical reasons, short-cuts and simplifications (form factors, effective power, etc) may be introduced, which then become an unexpected source of error.

To avoid this kind of error it is convenient to start burnup calculations with a fresh core and calculate in a systematic way all steps in the operating history, one after another. Sometimes this can be very impractical, because the core configuration could be changed hundreds of times since the fresh core. In such a case, the calculations can be started also with partly depleted core but should reach sufficiently far back into the operating history so as not to significantly influence the final result (present burnup).

At the end of each step of the calculation, the burnup of each fuel element is determined. It can now be used to compare with experimental results and to correlate with fission product buildup in a fuel element.

3. MEASUREMENTS OF REACTIVITY EFFECTS

The most simple method for determining the burnup of a single fuel element is based on the measurement of its reactivity effect in a core. The method combines the measurement of reactivity with the unit-cell burnup calculations, which are necessary to relate the measured reactivity to the burnup of the particular fuel element.

The fundamental parameter that connects the results of the calculations with the experimental data is the k_{∞} of each fuel element. The cross section code that generates the group constants (transport code) also calculates the k_{∞} as a function of burnup. For each type of fuel element, its k_{∞} depends only on its burnup and its value when fresh. The burnup dependence of k_{∞} is significantly different for different fuel element types, as shown in Fig. 5, using the FLIP, LEU, STANDARD- 8.5 w/o U and STANDARD- 12 w/o U TRIGA fuel elements as examples. Even fuel elements of the same type do not have precisely the same enriched inventory of uranium (or burnable poison) and, hence, their k_{∞} 's when fresh will vary slightly.

The k_{∞} of a fuel element is not measured directly. What is measured is its effect on the core reactivity.

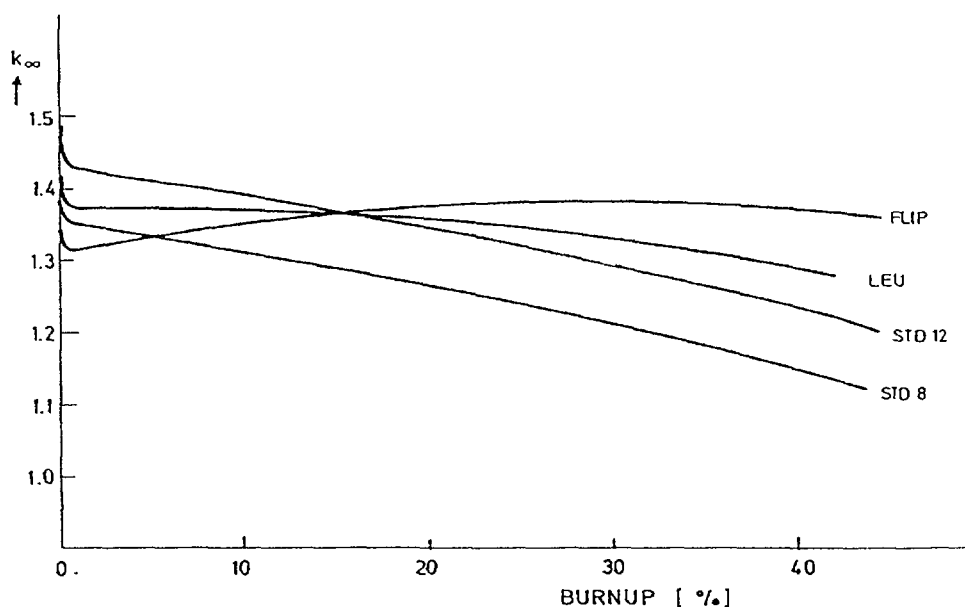


FIG. 5. Burnup dependence of k_{∞} for different fuel element types.

By assuming that the reactivity of a fuel element in the core (proportional to k_{∞}) is a known function of its burnup and by measuring the reactivity contribution of an element in a core, its burnup can be determined.

Fuel element reactivity is measured in a chosen core configuration wherein there is a single core position assigned for making the measurements. This core position is referred to as the experimental position, EP. A fuel element is placed in the EP and the reactivity (ρ) of the core measured. It is recommended to choose the EP near the core center (where the neutron importance is high) but distant from the control rod positions or irradiation channels so as not to disturb the flux distribution in the EP.

Usually a set of reactivity measurements for several fuel elements of the same type is performed. The burnup of at least one of them (reference element(s)) must be known, that is, determined by some other method. In special cases, the burnup of other elements can be determined without reactor calculations – simply by interpolation or extrapolation of the data. But in general reactor calculations are required. The choice of the method depends on the number of reference elements and on the dependence of k_{∞} (and reactivity) on burnup, which can be linear or non-linear.

a. Linear dependence of k_{∞} versus burnup

In this case burnup of measured elements can be determined without computer calculations, provided that the burn-up of at least two measured elements is known. The burnup of any other measured element can be obtained from its reactivity by interpolation/ extrapolation along the curve which is defined by the reference points (see Fig. 6). If

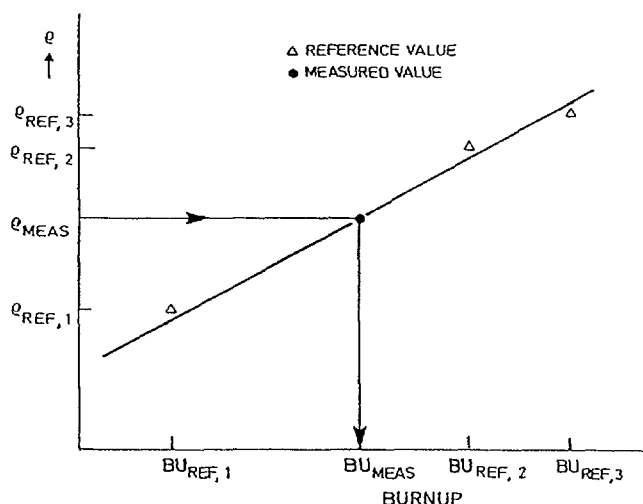


FIG. 6. Example of linear dependence of k_{∞} vs burnup.

there are more than two reference points, the interpolation line is constructed using the least square method.

b. Non-linear dependence of k_{∞} versus burnup

If k_{∞} for a given fuel element is a non-linear function of burnup then measured reactivity is also expected to be a non-linear function of burnup (e.g. FLIP, see Fig. 5). In this case more than two reference burnups are required for determining the interpolation function.

c. In both cases only one reference burnup is sufficient, if reactor calculations are applied. The procedure is as follows:

- core reactivity is measured with reference element (ρ_{ref}^M) and also with element with unknown burnup (ρ_M^M) in EP
- reactivity of the core with the reference element in EP is calculated (ρ_{ref}^C) using an appropriate computer code
- the burnup of the measured element is determined by trial and error method: in the first step, a guess is made for the burnup of the measured element and reactivity of the core configuration with this element in EP is calculated (ρ_M^C). The difference ($\rho_{\text{ref}}^C - \rho_M^C$) is calculated and compared to ($\rho_{\text{ref}}^M - \rho_M^M$). If the agreement is good (~ 1 cent), the guess for the burnup is the accepted value. If not, the procedure is repeated. In practice, not more than two or three guesses are necessary.

It is very convenient if a fresh element (burnup = 0) is available as the reference element because in this way the uncertainty in the reference burnup is avoided.

4. FISSION PRODUCT ANALYSIS THROUGH GAMMA RAY SPECTROSCOPY

In this section the techniques required to determine the burnup of a fuel element using gamma ray spectrometry to measure the fission product inventory in the fuel element is described. The first task to be performed to effect this method is to calculate the fission product inventory as a function of burnup of the fuel element as described below.

4.1. Calculations of the Fission Product Inventory

For isotopic composition and fission product inventory of irradiated fuel, special types of computer codes have been developed. These codes take into account detailed isotopic composition with decay chains for a large number of nuclides, while employing a relatively simple neutronics model (geometry of the system and energy dependence of cross sections and flux). Usually, zero-dimensional, one-group approximations are used, sometimes called point approximations. Several such point-depletion codes were developed for these purposes which are usually provided with its own library of few-group (usually one group) microscopic cross sections for a large number of isotopes. Generation of such a library is a complex process and must be performed with sophisticated transport codes which can take into account energy dependence of cross sections and the influence of neutron spectrum and self-shielding within the fuel element.

ORIGEN 2 [13], [16] is an example of a well known and widely available point-depletion code with a large number of one-group cross section data bases for different types of power reactors. However, its use for research reactors is limited because its data library was established only for power reactors. Composition of research reactor fuel is in general different from power reactor fuel. This is true also for the energy spectrum used for averaging the multigroup cross sections. Due to this, the code and the library should be tested before using it for research reactor calculations.

The applicability of ORIGEN 2 for TRIGA research reactors was tested by comparing results of depletion calculations (using the original PWR library) to WIMS-D/4 results for the most important isotopes. Three typical TRIGA fuel types were treated [15].

Because there were differences in isotopic concentration of actinides (which are the source of fission products) a new TRIGA specific library of cross-sections was prepared and introduced into ORIGEN2 as a function of burnup [14]. According to the ORIGEN 2 one-group point model they were averaged over the spectrum using the WIMS-D/4 code. Cross sections were generated separately for different fuel types, since their spectra are significantly different.

Use of a TRIGA specific cross section library significantly improves the results of ORIGEN 2 for all three types. Similar analysis should be performed for other types of research reactors and fuels and (if necessary) specific libraries can be similarly developed.

4.2. Gamma Spectrometry

Gamma-spectrometry allows a non-destructive determination of the fission and activation product content of spent reactor fuel [17]. The concentration of these products depends significantly on fuel parameters which describe the irradiation history and the fuel composition. Therefore, by performing gamma spectrometry of reactor fuel the burnup of this fuel can be determined experimentally. The possibilities and the basic limitation of deriving information from measured gamma spectra are discussed in this section. Further typical experimental arrangements for gamma scanning of reactor fuel are presented in this chapter.

Gamma ray techniques

Gamma rays emitted from spent fuel originate from fission products inside the fuel and from activation products of the cladding and other materials in the fuel (i.e. burnable poison, reflector material etc).

From the measured activity of a fission product isotope (see Appendix 8), A_{ij} , the total number of atoms, N_i^0 , present in the fuel element at the end of its exposure in the core can be calculated from the following relationship:

$$N_i^0 = \frac{A_{ij} \cdot \exp(\lambda_i t_c)}{\lambda_i} \quad (6)$$

- N_1^0 = number of atoms of i-th isotope at the end of irradiation exposure
 A_{ij} = measured activity of i-th isotope using the j-th energy gamma ray (s^{-1})
 λ_i = decay constant for the i-th isotope (s)
 t_c = time elapsed since the end of irradiation exposure and the measurement (usually the cooling time) (s)

The amount of a given nuclide in the fuel can also be calculated from the fuel history using the ORIGEN 2 code with the reactor specific neutron spectra as described in Section 4.1. The calculations made by ORIGEN 2 are relatively accurate, compared with the errors made in performing experiments.

Gamma ray spectrometry requires sensitive, sophisticated equipment in order to achieve reliable results. To obtain reasonable precision a number of peripheral effects have to be monitored during the measurement process and corresponding correction factors have to be fed into mathematical formulae.

The activity A_{ij} of a fission product i in a fuel element is extracted from the j-th energy peak (see Appendix 5) of a gamma spectrum by evaluating the following equation:

$$A_{ij} = \frac{P_j (1 + DTC) \cdot Z(t_m)}{B_{ij} \cdot EFF(E_j) \cdot ABS(E_j)} \cdot \frac{1}{t_m} \quad (7)$$

where

P_j = the net peak area under the j-th energy peak in question (most modern multichannel analyzers provide for background subtraction, live time correction, efficiency calibration etc.)

DTC = dead time correction*)

$EFF(E_j)$ = efficiency of the measurement set-up (comprises geometry factors and energy dependent detector efficiency as long as they are not part of the multichannel analyzer software)

t_m = measuring time (real time with DTC or live time without DTC) (s)

$ABS(E_j)$ = correction factor for the self-absorption of gamma quanta in fuel and cladding material for the j-th energy.

*) if there is no adequate on-line correction

$Z(t_m)$ = correction factor for the decay of the isotope during the measuring time (will be uniform in most cases)

B_{ij} = branching ratio; number of j-th energy gamma rays per disintegration - see gamma tables (Appendix 7).

The accuracy of the measurement is improved by making use of intensity ratios of gamma rays instead of absolute values. First of all this method minimizes the mathematical effort (although it still required an effective computer code) and at the same time it serves to stabilize the results of the data acquisition in terms of making them less sensitive to specific geometry and efficiency properties of several components of the set-up. For the ratio of the activities of two isotopes, i_1 and i_2 , the above equation is simplified to:

$$\frac{A_{i1j1}}{A_{i2j2}} = \frac{P_{j1}}{P_{j2}} \cdot \frac{B_{i2j2}}{B_{i1j1}} \cdot \frac{EFF(E_{j2})}{EFF(E_{j1})} \cdot \frac{ABS(E_{j2})}{ABS(E_{j1})} \quad (8)$$

When determining the ratio of two lines with $E_{j1} \approx E_{j2}$ the factors EFF and ABS may be omitted.

Furthermore a few of these activity ratios can be related to certain variables of the irradiation history, i.e. cooling time, irradiation time, integral flux. The combination of these calculations and the existing operational data is then used to perform the iteration routine to produce the burnup parameters.

The number of ratios that can be used independently for this purpose is limited in principle and by measurement difficulties. For each variable of the irradiation history a different set of ratios is used which is qualified by its low dependence on other variables. From this set the ratios with the most convenient accessibility for the measurement process under the given circumstances (derivation of irradiation time, derivation of cooling time) are selected [19].

This is illustrated with the following three ratios which are useful for burnup determination:

Cs-134/Cs-137	very independent of other parameters apart from the neutron spectrum. Cs-134 can only be seen after a cooling time of 1-2 months.
---------------	---

- | | |
|----------------|--|
| Eu-154/Cs-137 | not as linear with the integrated flux as Cs-134/Cs-137, but might show less dependence on neutron spectrum. Eu-154 can only be seen after a cooling time of approximately 1/2 year. |
| Cs-134/ Sb-125 | Sb-125 can only be seen after a cooling time of approximately 1.5 years, but shows even less dependence on irradiation time than Cs-134/Cs-137. |

The following ratios can be used to determine the cooling time:

- Zr(Nb)-97 / Te(I)-132
- I-133 / Te(I)-132
- Te(I)-132 / Ba-140
- I-131 / Ba(La)-140
- I-131 / Ce-141
- Ba(La)-140 / Ce-141
- Ba(La)-140 / Zr-95
- Ce-141 / Zr-95
- Ce-141 / Ce-144
- Zr-95 / Nb-95
- Ce-144 / Cs-137

and similarly for the irradiation time the ratios

- Ba(La)-140 / Cs-137
- Zr-95 / Cs-137
- Ce-133 / Cs-137

are useful although in these cases, every ratio depends also on other variables (isotopes in brackets are daughter isotopes and may also be used).

Today Ge-HP detectors or Ge(Li) detectors are the most important tools for gamma spectrometry of spent reactor fuel. However before evaluating the results from these instruments a number of correction factors have to be taken into consideration, such as the detector efficiency, the geometry factor, scanning errors, dead-time correction, self-shielding effects, decay time, etc.

Efficiency

Assuming a point γ -source at distance r from a detector, only a fraction $F/4\pi r^2$ (F = sensitive detector cross section with the collimated beam area) is seen by the detector (geometric efficiency). In addition the

detector itself has an efficiency which depends on the γ -energy (detector efficiency). This detector efficiency has to be determined prior to the measurement by using calibration γ -sources (see Appendix 6) of known activity (Eu-152 is recommended) [18]. Most of the γ -spectroscopy software offers reliable support for efficient calibration. The geometric efficiency of the experimental arrangement has also to be determined by placing (strong) calibration sources into the given arrangement (see therefore Figure 7).

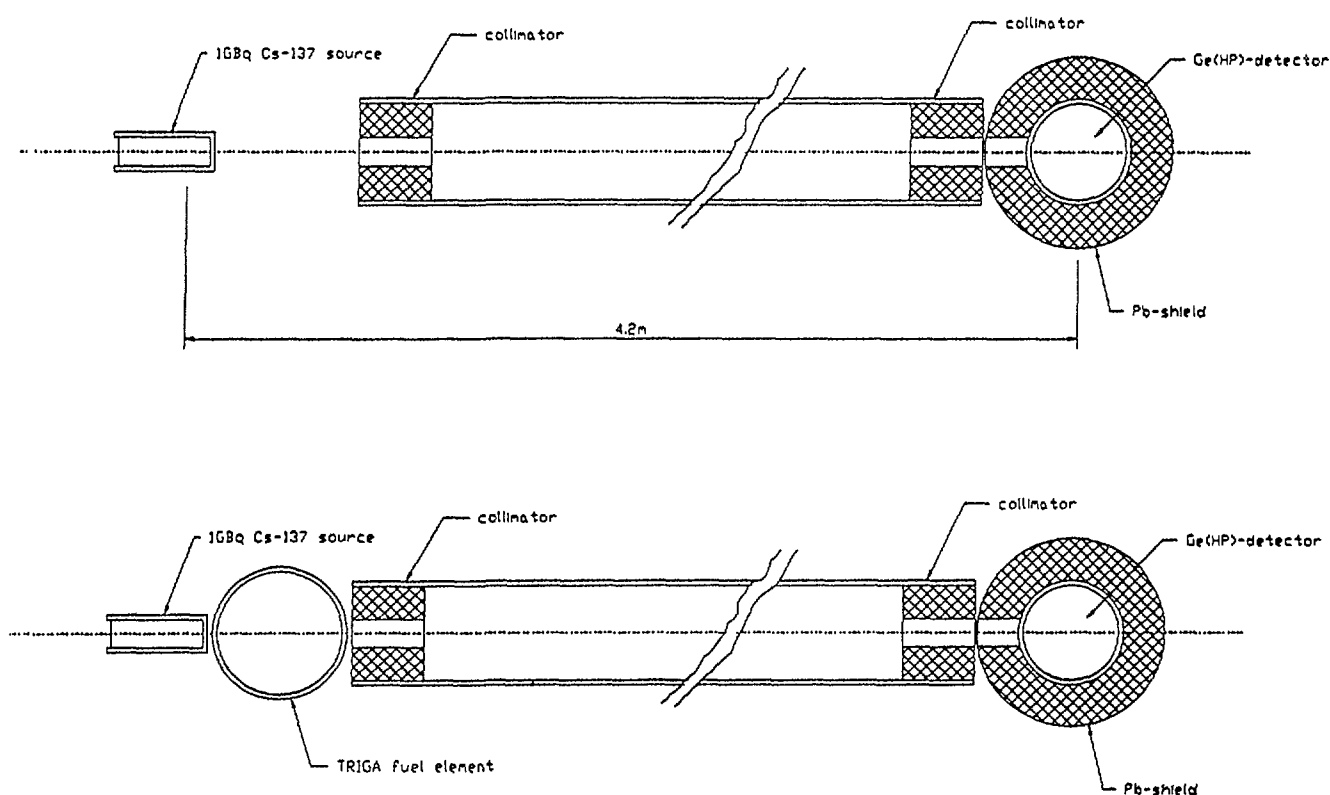


FIG. 7. Calibration setup.

Self-shielding effects: It has to be emphasized that this calibration procedure includes errors due to the following reason: Calibration sources (i.e. Cs-137) are usually point sources compared to spent reactor fuel. The γ -rays from the spent fuel represents a volume source of more or less homogeneous distribution in the fuel and is therefore subjected to self-shielding effects from within the fuel. The self-shielding factor depends on the fuel material composition, the geometrical behaviour, and on the γ -ray energy. An optimal calibration source would be a piece of fuel made of the same material and size as the investigated fuel with a known burnup (Fig. 7).

Scanning errors: Usually the spent fuel is positioned several meters under water in a scanning device. The scanning collimator has to be moved remotely from the reactor pool top. It is therefore very important that the position of the scanning collimator can be reproduced accurately. With a typical water depth of about 5 meters, a position accuracy of the scanning collimator of about 1 to 2 mm can be achieved.

Dead-time corrections: Depending on the experimental arrangement (distance of fuel element to the detector, collimator, activity of the fuel) errors may be introduced by high count rates due to pulse pile-up. During processing of high count rates a detector and its associated electronic equipment is not able to detect all incoming signals, because it is "too busy" in handling all the incoming information. Therefore, during a certain time (= dead time) the detector does not accept signals. If this dead time is high (> 3%) then the results have to be corrected. Modern instruments usually indicate the actual dead time and offer an adequate dead time correction (live time).

Decay time: When a fuel element is removed from the reactor core the fission and activation products decay according to their half-lives. For short cooling periods and long half life (such as Cs-137) the decay time is not important, but if nuclides with half lives in the range of the cooling time t_c are investigated the decay time has to be taken into account. Therefore, the simple decay time correction is strictly recommended. Decay time correction during the measuring time t_m can be omitted in most cases.

Collimator: As mentioned above good collimation is important for γ -ray spectrometry of spent fuel. With underwater scanning devices such air-filled collimator tubes typically have a length of several meters. In addition, lead plugs with about 5 mm collimator diameter can be inserted into the collimator tube. Shielding the detector helps to increase the resolution and to reduce the detector dead time. A typical experimental set up for under water TRIGA fuel scanning is shown in Fig. 8. The advantage of an in-pool arrangement is that the measurements can be done in the reactor pool without removing the fuel from the pool.

Using special designed containers, fuel scanning can also be performed outside the reactor pool. A typical arrangement for this case is shown in Figs. 9 and 10. With these out of pool arrangements, the spent fuel is transferred from the reactor tank to a lead container by a fuel transfer cask. The fuel element is then lowered into the fuel scanning container.

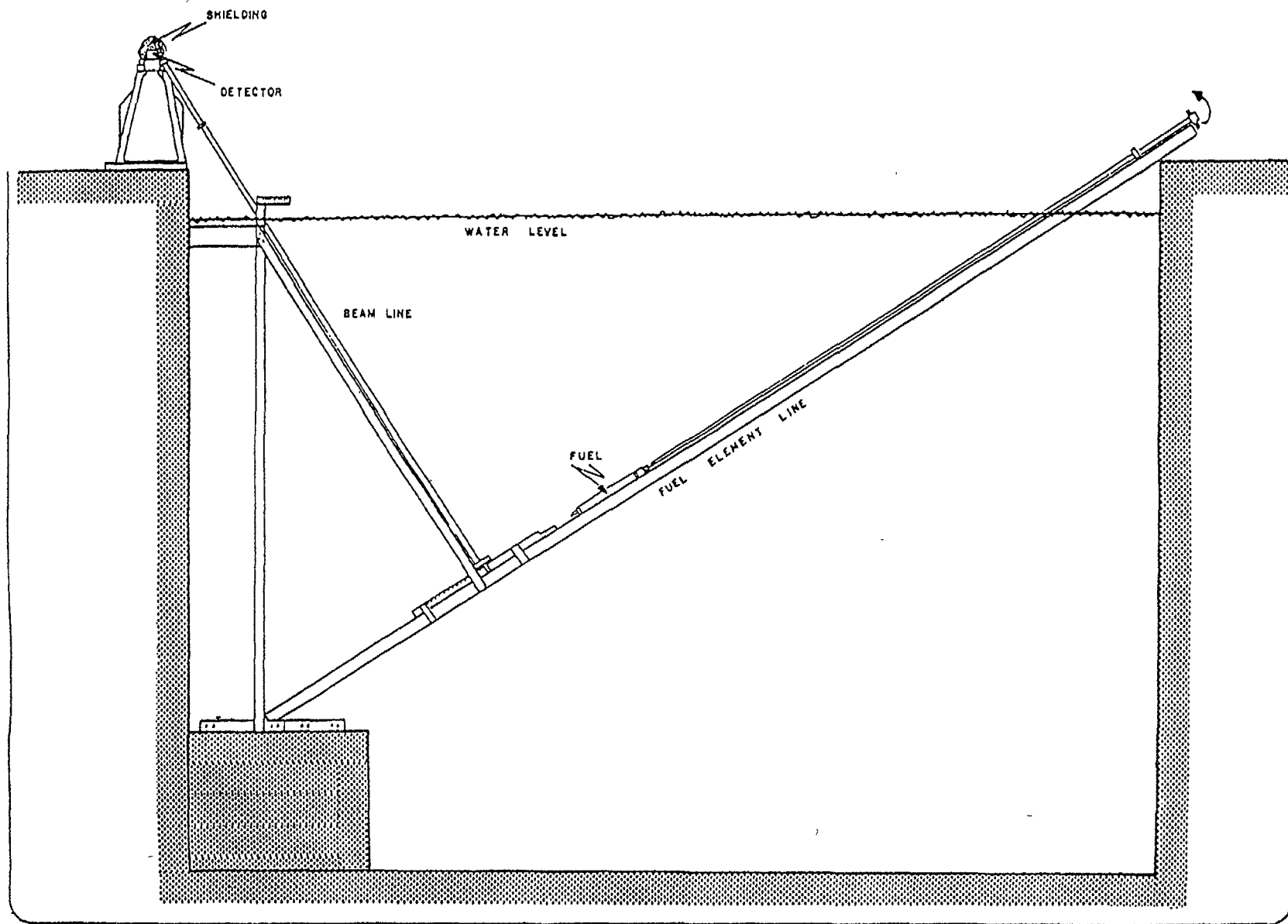


FIG. 8. TRIGA Mark III, Mexico — in-pool arrangement.

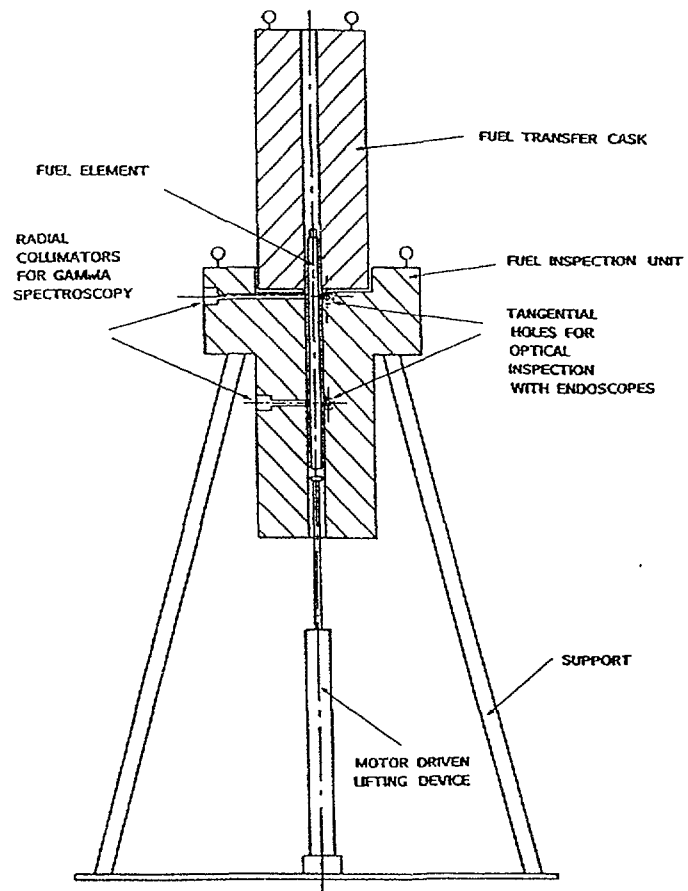


FIG. 9. Fuel element scanning device.

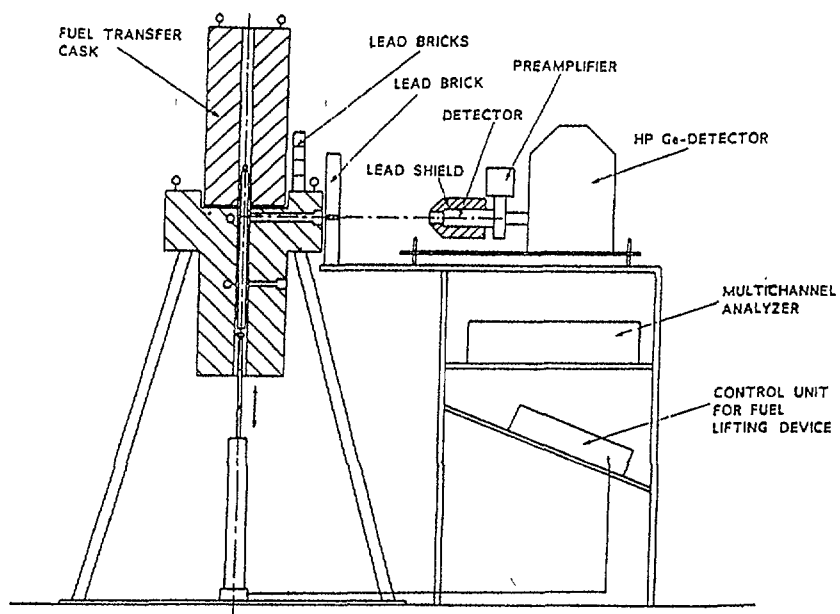


FIG. 10. Fuel element scanning device.

With the help of an electric piston the fuel element can be moved up or downwards in front of several collimator holes. The detector system is positioned in front of the collimator hole with a distance of about 1 m between fuel element and detector. Efficient shielding and good, reproducible alignment of the detector is important. The advantage of the out of pool measurements is the low background.

In-Pool Experimental Set-up for Gamma Ray Spectrometry

The experiment performed in the pool is designed for measuring the fission products in an irradiated TRIGA fuel element.

An experimental set-up for performing these measurements on irradiated TRIGA is shown in Fig. 8. The experiment is designed to allow sliding the fuel element to fixed positions along its length in ten 38 mm steps. This permits determination of the axial variation of the burnup in the fuel element (buckling). In addition, the fuel element can be rotated in each position to determine the effect of the radial gradients in the burnup distribution of the fuel. The tool that rotates the fuel element is able to indicate the rotation angle.

A sketch of the collimator tube is shown in Fig. 7. The collimator length is about 4.0 m and the inner diameter of the collimator is 5 mm. It can be easily enlarged, if the count rate at the detector is too low for reasonable statistical errors. The volume within the fuel which emit gamma rays to the detector is also shown in that figure (cylindrical source). Gamma rays from different points of the fuel element within the range of the view of the detector are attenuated differently according to their origin.

To determine the self-absorption of the gamma rays in the cylindrical source the following equation is used:

$$J = J_0 \cdot \exp(-\mu_f \cdot \alpha) \quad (9)$$

where

J_0 = intensity of the original beam

J = intensity of the attenuated beam

μ_f = linear attenuation coefficient (cm^{-1})

α = length of the attenuating material (cm)

The calibration procedure has to be made in two steps: One step is the detector efficiency calibration using a set of standard γ -sources covering the range of γ -energies to be measured. The other step is the geometrical factor which can be measured by using a strong γ source (e.g. Cs-137, 0.9 GBq) placed at the fuel element position on the collimator tube. For the arrangement in Fig. 5 with the above γ -source, an efficiency of 11.9 counts/s at 662 keV could be obtained. Together with a Ge-HP detector ($\phi 45$ mm, 42 mm long) and appropriate electronics the efficiency function EFF(E) was found as in Table 2.

TABLE 2. EFFICIENCY CALIBRATION
FOR AN ARRANGEMENT AS IN FIG. 7

<u>Energy</u> [keV]	<u>EFF(E)</u> [s ⁻¹ GBq ⁻¹]
121.8	24.802
244.7	18.170
344.3	15.659
605.0	12.312
662.3	11.853
688.7	11.661
778.9	11.077
796.0	10.978
867.4	10.594
964.0	10.142
1005.1	9.970
1085.8	9.659
1089.7	9.645
1112.1	9.565
1212.9	9.232
1299.1	8.978
1365.0	8.800
1408.0	8.690
1457.6	8.570
1850.0	7.789
2188.0	7.289

The linear attenuation coefficient μ_f of TRIGA fuel can be determined by using a fresh fuel element and a strong Cs-137 source together with the collimator tube and the detector in the set-up procedure. A linear attenuation $\mu_f(662 \text{ keV}) = 0,436 \text{ cm}^{-1}$ was found for a stainless steel clad LEU-Standard-TRIGA fuel element (cladding smeared into the fuel; no scattering taken into account).

In order to determine the fission product activity in the fuel element, the self-absorption can be taken into consideration using the following definition:

$$\frac{1}{\text{ABS}(E_j)} = \frac{1 - \exp(-\mu_f(E_j)\alpha)}{\mu_f \cdot \alpha} \quad (10)$$

where

$\mu_f(E_j)$ = linear attenuation coefficient of the fuel material
(cm^{-1}) for j-th energy

α = length of the cylindrical source (cm)

For the stainless steel clad standard TRIGA fuel element a self-absorption correction $\text{ABS}(662 \text{ keV}) = 0.484$ was found for Cs-137 experimentally.

The specific activity (activity per volume-unit) of the i-th isotope in that part of the fuel, which is under investigation can be found as:

$$A_{ij,\text{spec}} = \frac{I_{j,\text{abs}}}{B_{ij} \cdot V_s} \cdot \frac{1}{\text{EFF}(E_j)} \cdot \frac{1 - \exp(\mu_f(E_j) \cdot \alpha)}{\mu_f(E_j) \cdot \alpha} \quad (11)$$

where

V_s = volume of fuel "seen" by the detector

$I_{j,\text{abs}}$ = net count rate (s^{-1}) for the j-th energy peak (background subtracted, dead time corrected, ...)

$A_{ij,\text{spec}}$ = specific activity (s^{-1}) of the i-th isotope using the j-th gamma energy photo peak

$\text{EFF}(E_j)$ = detector efficiency for the j-th energy

The number of gained nuclides of the i-th isotope per volume unit by fission is then

$$N_{i,spec} = \frac{A_{ij,spec} \cdot e^{\lambda_i t_c}}{\lambda_i} \quad (12)$$

The number of fissioned (burned) U-235 atoms per volume unit ($N_{b,spec}$) is related to the fission yield for the isotope used so that

$$N_{U-235,spec}(initially) - N_{U-235,spec}(currently) = N_{b,spec} = \frac{N_{i,spec}}{\gamma_i} \quad (13)$$

where

$N_{i,spec}$ = number of nuclides of the i-th isotope per volume unit

γ_i = fission yield for the i-th isotope related to U-235 (see Appendix 9).

The burnup of the fuel investigated according to the FIFA (fissions per initial fissile atom) definition is:

$$FIFA = \frac{N_{U-235,spec}(currently)}{N_{U-235,spec}(initially)} = 1 - \frac{N_{i,spec}}{N_{U-235,spec}(initially)} \cdot \frac{1}{\gamma_i} \quad (14)$$

For operating cycles as usual at research reactors, an additional correction is recommended: the burnup decay in addition to the cooling time decay.

If the half life is long compared to the cooling time as it is usually with Cs-137, the specific activity is directly related to the total fission density, and the burnup decay factor (BUD) may be written as

$$BUD = \frac{\sum_{k=1}^m t_{ok}}{\sum_{k=1}^m t_{dk} \cdot \exp(\lambda_i \cdot t_{dk})} \cdot \frac{1}{\gamma_i \cdot \lambda_i} \quad (15)$$

where

m = number of the reactor operation cycle during which the fuel element was irradiated

t_{ok} = operation time of the k -th operation (s)

t_{dk} = time from the k -th reactor shutdown until the activity (s) is counted

λ_i = decay constant for the i -th isotope (s^{-1})

γ_i = the fission yield for the i -th isotope.

Gamma Spectrometry Outside of the Reactor Pool

For this purpose a measurement device was designed and constructed at the Atominstitut der Oesterreichischen Universitaeten Wien. It is composed of a massive lead container for fuel elements and a collimation system which points to the necessary electronic equipment, i.e. Ge-HP detector and multichannel analyzer. This container is situated outside of the reactor pool in a corner of the reactor hall. Fuel elements can be transferred with the fuel element transfer cask, either from the spent fuel storage or directly from the reactor core after having cooled down for some time. Once inside the fuel scanning device, every part of the element can be positioned in front of the collimator system using a remote controlled electrical lifting device. It is therefore possible to scan the element with a resolution of 1 mm as well as to rotate the element inside of the container to scan it around its vertical axis.

The fuel element scanning device (Figs. 9 and 10)

The fuel element scanning device is primarily a lead container with an axial hole into which a standard TRIGA fuel element fits loosely so that it can be moved safely. This lead container is held at a level of about 1.8 m above the ground by a sturdy steel construction. The axial hole for the fuel element penetrates all the way through the container. At its lower end is the top cylinder of an electrical lifting device which is positioned below the lead container on the ground and reaches up to the bottom of the fuel element, thereby holding it inside the container. This electrical device is remotely controlled from nearby. Presently it has a reproducibility of position of 1 mm. The lead shield has a minimum thickness of 200 mm, thereby guaranteeing a maximum dose rate of 2.5 mR/h on the outer surface of the shield, as long as the fuel element has been out of the core for at least 14 days prior to insertion. The lead shield has totally 7 collimator openings at different

positions leading from the inner axial hole to the outside. They all have a diameter of 10 mm; 3 of them lead radially from the center to the outside, the other 4 are tangential holes which traverse the whole container. The tangential holes are used primarily for optical inspection. The experiments for this work were carried out at one of the radial holes.

An additional collimator was inserted into this collimator hole (described below). At a distance of 540 mm from the surface of the lead shield, that is 970 mm from the surface of the fuel element, the Ge-HP detector was positioned.

The collimator

This is stepped lead cylinder with an outer diameter of 10 mm and a 5 mm collimator hole which fits tightly into one of the radial openings in the lead container. An optimal stepped collimator as described above will both eliminate scattered γ -rays and will result in an acceptable resolution of the γ spectrum. In addition to the collimator, a 40 mm thick lead brick with a 5 mm hole was placed between the collimator and the detector and its hole aligned with the collimator.

This resulted in a stable collimated beam with a length of 310 mm and a diameter of ≤ 5 mm. The whole system was finally aligned using a laser, assuring that the HP-Ge detector was positioned properly so that the radiation beam impacts in the center of the sensitive detector zone. In the experiments, the distance between the surface of the fuel element and the sensitive zone of the detector was 970 mm. At the multichannel analyzer the dead time was always $\leq 10\%$, typically 5-7%, less for fuel elements with long cooling times.

Experiments inside the reactor pool

A fuel scanning device for measurements inside the reactor pool also exists at the Atominstitut der Oesterreichischen Universitaeten Wien (see Fig. 11). This is used for in-pool experiments. It consists of a metal tube into which a fuel element fits tightly. The tube is lowered onto the top of the thermal column inside the reactor pool and a fuel element is inserted. The Ge-HP detector is placed vertically above the tube outside of the pool and is connected with it by an air-filled collimator tube. The collimator tube can be moved horizontally from above by means of a long metal drive shaft. The

detector has to be adjusted accordingly. In this way, a fuel element can be scanned underwater [20].

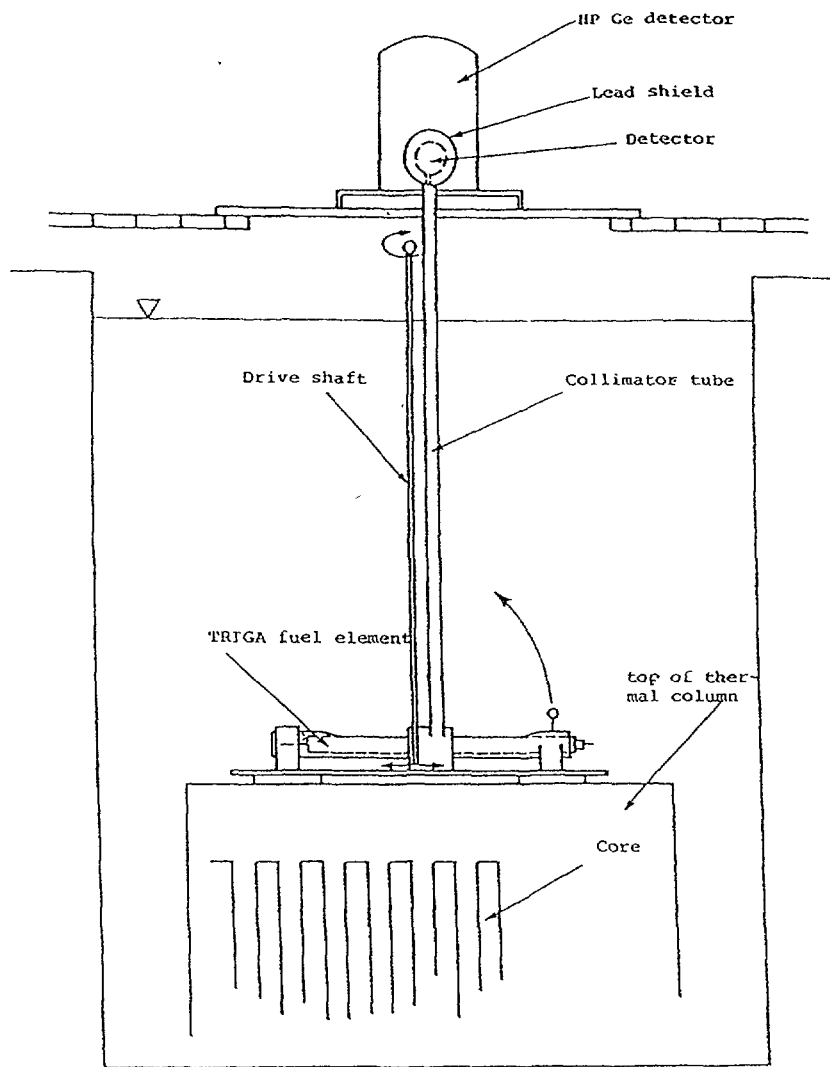


FIG. 11. In-pool fuel element scanning device.

In-Pool Scanning Devices

Several in-pool scanning devices for MTR fuel elements are known. One type of construction is designed to see the whole full element and collimate only at the detector in order to reduce the count rate to achieve reasonable dead times [22]. Another type of construction only guides a fine collimated X-ray beam to the detector. Therewith it is easily possible to scan the fuel element in order to obtain the shape of the burnup in the vertical direction. The advantage of such in-pool determinations for the fuel burnup is that the fuel remains in the pool and therefore no radiation risk as well as no cooling problems (decay heat) occur. A typical arrangement of an in-pool scanning tool for MTR fuel is shown in Fig. 12.

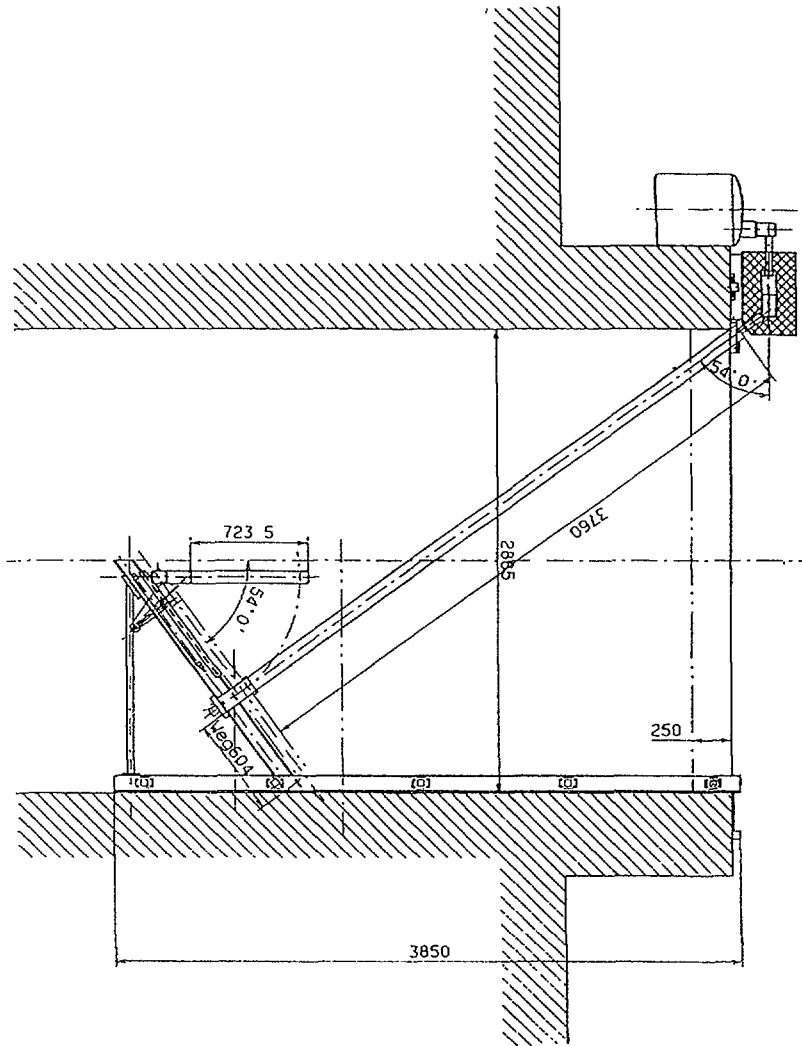


FIG. 12. Saphir arrangement for MTR type burnup determination by gamma scanning.

Appendix 1

DETERMINATION OF MTR FUEL BURNUP USING THE REACTIVITY METHOD

The method is used at the 10 MW MTR reactor at the Paul Scherrer Institute in Villigen, Switzerland.

The aim of this measurement is to determine the burnup of each fuel element calculated according to its operating history and to make corrections if obvious differences result. This helps first, to build reliable new operating core configurations and second, economizes the fuel consumption in order to reach high burnup (up to 70%) with homogeneous flux distributions in the reactor core.

In order to have the optimal parameters for this investigation, it is necessary to build a special core configuration. This core configuration contains no beryllium-reflector and consists only of fresh 93% enriched fuel in the most compact arrangement for criticality (cold, unpoisoned). A beryllium-reflector element serves as a neutron source with low coupling to the core. The reactivity determination is one of the important procedures during this determination. Therefore, a rather sophisticated computer-controlled rod-drop-method combined with a high count rate neutron channel is used to compare the excess reactivity of core loadings with some well known burnup standard elements to that of the unknown fuel elements investigated. As only one shim rod is used to compensate for this excess reactivity, the initial loading has to be chosen in a way that for a fresh fuel element the rod is half in, while for a spent fuel element, in the measuring position, that rod is not completely out in order to get the reactor critical. Such a core configuration is shown in Fig. 13.

A typical schedule for a single reactivity measurement in a proper adjusted core is:

1. Insert the fuel element investigated (verify the fuel element number).
2. Run the reactor in a critical status at 1kW_{th} in automatic mode at constant temperature (forced cooling).

	5	6	7	8	9
3	330 90%	323 90%	331 90%	345 93%	
	GR1 0%	FE 8%	GR2 0%	FE 1%	NS
4	343 93%	342 93%	348 93%	305 90%	
	FE 1%	FE 0%	FE 0%	FR 40%	
5	332 90%	344 93%	333 90%		
	GR3 1%	FE 0%	GR4 1%		
6	346 93%		347 93%	341 93%	
	FE 1%	MP	FE 1%	FE 4%	FC

SAPHIR Core Configuration LDG 576
(Clockwise from upper left, for each grid position, are
given: El.No., enrichment, burnup and description)

Description:

FE standard MTR fuel element
GRn main control element, n=1,2,3,4
FR regulating element (fine)
NS neutron source
MP measurement position
FC fission chamber

FIG. 13. Saphir core configuration.

Conditions: 3 of 4 shim rods near the measuring position fully withdrawn, regulating rod as far out as acceptable (e.g. around 80% out). Try to always reach a similar position for subsequent measurements in order to have only a few corrections for the differences in regulating rod positions.

3. Wait at least 15 minutes for the reactor period to stabilize.
4. Run the reactor without automatic control to check the stability (two to five minutes).
5. Start rod drop programme - reactor shutdown.
6. Save and verify the reactivity data taken; note element number, shim rod position, and reg rod position.
7. Exchange the investigated fuel element with another fuel element.
8. Run the reactor as under 2 and so on.

The gained reactivity values (compensated by the regulating rod) of one element to another contain the burnup information respectively. If the fuel elements are identical as it is for the same type of fuel, a linear correlation between the measured reactivity and the burnup can be assumed.

For scaling, it is necessary to have at least two elements with a known but different burnup; a fresh fuel (0% burnup) and a spent element (for example 50% burnup). This burnup standards have to be measured several times in order to check the reproducibility of the whole method.

The schedule for a measurement cluster of reactivity measurement could be:

1. burnup standard (0% burnup)
2. unknown element
3. unknown element
4. burnup standard (50%)
5. unknown element
6. unknown element
7. burnup standard (0% burnup)
8. unknown element
9. unknown element
10. burnup standard (50% burnup)
11. unknown element
- ... and so on.

For such a measurement operation cycle, one has to have sufficient operation time. Otherwise the accuracy of the determination which is in the range of 3 to 5% drops off significantly, mostly because of lack of stability of the critical reactor.

In addition, it has to be kept in mind that only identical fuel elements can be compared for reactivity and for their burnup. If there is a different design (like it is for shim rod elements and for regulating rod elements) or if the fuel meat composition is different (as it is for 20% enriched LEU elements and for 45% enriched MEU elements), corrections or better appropriate standards have to be used to scale the reactivity measurements to their burnup.

Appendix 2

DETERMINATION OF FUEL ELEMENTS BURNUP BY REACTIVITY MEASUREMENTS AT TRIGA REACTORS

This method is used at the TRIGA Mark II reactor at the Atominstitut in Vienna, Austria.

1. Preparation for the measurements

- a. A proper experimental position (called EP) has to be chosen in the core in order to avoid geometric influence on reactivity:
 - The position should be in a ring with high importance, since reactivity changes to be measured are comparatively small.
 - The EP should not be close to control rods or irradiation channels because a large neutron flux deformation can be expected there. A homogeneous core (i.e. core with only one type of elements) would be an ideal case.
- b. Reactor poisons as Xe or Sm have a great influence on reactivity. It is therefore important to have a reactor which is Xe-free and at Sm equilibrium.
- c. Temperature has a great influence on reactivity too, so a comparatively low thermal reactor power (e.g. 10 W) should be chosen to avoid temperature gradients and the variations during the measurements.
- d. Reactivity changes at a given burnup vary significantly depending on the type of element, therefore different measurements for different types (e.g. Al or SS clad elements, FLIP elements) have to be carried out. Even the slightly different U-235 content of different fuel elements must be taken into account (check the shipping documents), especially when only small burnup values are expected.

2. Carrying out the measurements

- a. The procedure starts with the removal of all elements to be investigated from the reactor core. They should be placed in storage racks.
- b. A reference element has to be placed in the experimental position and the control rods have to be withdrawn to set the reactor just critical; the positions of the rods must be noted since the difference is a measure of the reactivity change for the fuel investigated compared to the standard fuel element.
- c. The reactor has to be made subcritical. The reference element must be removed and the element to be investigated should be placed in the experimental position. The control rods have to be raised again until criticality is reached. The positions of the rods must be noted. Sufficient time must be allowed for the reactor period to stabilize.
- d. This procedure has to be carried out successively with all elements to be investigated.
- e. It is recommended to repeat at least a few measurements to see if the results are reproducible. Moreover, some elements should be turned around their axis in the EP to see if they have homogeneous burnup.
- f. A calibration of the control rod (e.g. by the reactor period method) should be done at the beginning and at the end of the measurements in order to determine Δk from the differences in rod positions, since it is in direct relation to burnup.

Appendix 3

DETERMINATION OF FUEL BURNUP BY REACTIVITY MEASUREMENTS AND REACTOR CALCULATIONS

Introduction

This is a description of a method for determining fuel element burnup which was developed at the 250 kW TRIGA Mark II reactor at Jozef Stefan Institute in Ljubljana. It has been in practical use since 1985.

Compared to other similar techniques, this method has two advantages: it is insensitive to control rod position (control rods are completely withdrawn during the measurement) and to control rods worths measurements (reactivity is measured directly from reactor period measurement and not from control rod worth).

Description of the Method

Before starting the experiment the reactor must be shut down for at least 50 hours to eliminate xenon effects. All reactivity measurements are performed at zero power (~ 100 W) to avoid temperature effects. All elements to be measured (reference + elements with unknown burnup) must be removed from the core and EP defined. C-8 location is usually used because it is near the center of the core and it is easily accessible.

It is convenient if the experiment is prepared in such a way that the core reactivity is always positive and the doubling time method can be used. For this reason the reactivity of all measured elements is roughly estimated and the experiment is started with the least reactive fuel element in the EP.

The experimental procedure is as follows:

- First element to be measured is inserted in EP and the critical core configuration is established by adding or removing elements in outer ring. In order to improve experimental conditions and to enhance calculational analysis compact and cylindrical geometry should be approached as much as possible. As in practical situation

reactor cannot be made exactly critical, the reactivity must be measured, but it should not exceed few cents (~ 20 pcm) to enable this (and other) reactivity measurement with fully withdrawn control rods.

- Shut down the reactor and replace the element in C-8 with next one to be measured. Achieve criticality by withdrawing the control rods. If the elements to be measured were arranged according to increasing reactivity, the reactor will be (slightly) supercritical in this step. To the contrary, the reactor can be subcritical and the measurement is not possible. From the position of the control rods estimate the core reactivity and make sure that the corresponding reactor period will not exceed the SCRAM limit (in our case ~ 7 s – which corresponds to 0.4 \$ of inserted reactivity). If so, withdraw the control rods completely and measure the reactivity using doubling time method.

If the period is too short to use this method read the reactivity value from the control rod position. However, it is recommended to avoid this situation because the accuracy of reactivity measurement is much lower. In such a case it is better to move the EP to the outer rings to decrease the reactivity effects and repeat the whole set of measurements.

It is important that all elements are measured at the same position and under the same conditions (temperature, power level, core configuration). Otherwise the results are not directly comparable.

Evaluation of experimental results

Evaluation of experimental results depends on the number of reference burnup values:

- a. Two (or more) reference burnups

Reactor calculations are not necessary. Determine the unknown burnups by interpolation.

Make a diagram of reactivity values of reference elements as a function of burnup and fit a curve through them. Note that the shape of the curve is proportional to k_{∞} versus burnup diagram,

presented in Fig. 5 of the main part of this report. If the burnup interval of measurement is small, even non-linear curves can be approximated linearly.

b. One reference burnup

Reactor calculations cannot be avoided. Use the TRIGAP computer code.

- First calculate the core reactivity with the reference element in EP ($\rho_{\text{ref}}^{\text{C}}$).
- Make an initial guess for burnup of the first element to be calculated.
- Insert the estimated burnup value into the fuel element data file (ELEM.DAT)
- Calculate the reactivity of the core with this element in EP ($\rho_{\text{M}}^{\text{C}}$).
- Compare reactivity difference ($\rho_{\text{ref}}^{\text{C}} - \rho_{\text{M}}^{\text{C}}$) with corresponding measured values. In case of agreement, the estimated burnup is the final result. If there is no agreement a new guess is made for burnup and the procedure repeated.

Note: If the EP is properly chosen and the burnup differences are not big, the measured reactivity differences do not exceed 100 pcm. For this reason it is important that reactivity measurements are carefully performed (use of reactivity meter recommended!) and analyzed.

Appendix 4

EXPERIMENTAL PROCEDURE FOR MEASURING THE REACTIVITY (k_{∞}) OF FUEL ELEMENTS

This method was developed and has been in current use at PSRR TRIGA Mark II reactor at Pennsylvania State University, Pennsylvania, USA.

In these experiments a configuration is chosen wherein there is simple core position to be used as the experimental position, EP. The fuel elements are placed sequentially in the EP and the corresponding ρ of the core measured. The EP is chosen to be near the core center but distant from the calibrated control rod position so as not to disturb the flux in the EP.

Referring to Fig. 1 of the report, the core position I-10 is used as the EP, and in position F-9 is the safety rod (SA) which is the calibrated control rod. The ideal configuration is one wherein the core is slightly supercritical with ALL control rods out of the core, i.e. at their full out upper limits, and when the least reactive (lowest k_{∞}) fuel element is in the EP. For the experiments in reference, the initial core measurement began with the unused 8.5 w/o fuel element in the EP core configuration having approximately 40 cents excess reactivity.

The control rod used as the calibrating rod can be accurately calibrated while performing the measurements. Period measurements provide an accurate means for calibrating the rod. It is important that the neutron source, used to start up the reactor, be removed from the core when making the period measurements. It is also important not to move the calibrated control rod to reduce the power after each period measurements.

The control rod to be calibrated is first adjusted to where the core is just critical with all other control rods at their full out position. Then with this position recorded for $\rho \approx 0$, one of the other control rods is selected for lowering the power to prepare for a period measurement. While the core is subcritical and at low power, the rod being calibrated is raised to a new position for a faster period and the remaining rods placed at their upper positions to measure the new period. A few (usually two) period measurements are made with the rod being calibrated at a fixed position to determined reproducibility. When the calibrated rod is raised to a position

which produces a period that is too fast to measure with all other control rods at their upper limit, then a second rod is moved into the core (do not use the regulating rod in position H-12) to where the reactor is just critical as in the first step. The procedure described above is then repeated as before except that the two rods used for calibration are not used for changing the power level for repeat period measurements.

Once the calibrated rod is placed where the period is too fast again, the second rod is moved slightly inward to bring the reactor just critical as before. These measurements are continued until the control rod is calibrated to cover the range in reactivities produced by exchanging the different fuel elements in the EP.

In performing the actual experiment, beginning with the unused 8.5 wt % U fuel element, it is best to set the calibrated rod for approximately a 30 second period and leave it there while the other control rods are lowered and the source reinserted into the core to allow replacement of the standard fuel element with an irradiated fuel element. When the measurement is made with this fuel element in this core it should be possible to go critical without changing the position of the calibrated rod but having the other control rods at their full out position. The period should be slower indicating a lower k_{∞} of the irradiated fuel element. The difference in reactivity of the core is determined in the most accurate way when the measurement is performed in this manner.

If the calibrated rod must be moved to bring the reactor critical, the error in the measurement is increased by the error introduced from the error in the repositioning the control rod. This additional error can be made small by always setting the calibrated rod on moving it in the same direction to eliminate any slack in the control rod positioning mechanism.

It is always good to repeat the measurement of the unused fuel element and a few other fuel elements to determine the accuracy of the measurements.

Appendix 5

FISSION PRODUCT PRODUCTION IN LWR FUEL ASSEMBLIES

MEASURABLE GAMMA-EMITTING FISSION PRODUCTS MEASURABLE FISSION PRODUCTS IN FUEL ELEMENTS

<u>Isotope</u>	<u>Half-life</u>	<u>Principal Gamma Rays</u> (keV)
^{95}Nb	34.97 ± 0.03 days	765.8 (99.82%)
^{103}Ru	39.35 ± 0.05 days	497.1 (86.4%); 610.3 (5.4%)
^{95}Zr	63.98 ± 0.06 days	724.2 (43.1%); 756.7 (54.6%)
^{144}Ce	284.5 ± 1.0 days	696.4 (1.34%); 1489.2 (0.26%); 2185.6 (0.66%). [Gamma rays from ^{144}Pr ($t_{1/2} = 17.3$ min) daughter].
^{106}Ru	366.4 days	622.2 (9.8%); 1050.5 (1.6%); 1562.2 (0.17%). [Gamma rays from ^{106}Rh ($t_{1/2} = 29.8$ s) daughter].
^{134}Cs	2.062 ± 0.005 year	604.7 (97.6%); 795.8 (85.4%); 801.8 (8.7%); 1038.5 (1.00%); 1167.9 (1.81%); 1365.1 (3.04%)
^{154}Eu	8.5 ± 0.5 year	996.3 (10.3%); 1004.8 (17.4%); 1274.4% (35.5%)
^{137}Cs	30.17 ± 0.03 year	661.6 (89.9%); [Gamma ray from $^{137\text{m}}\text{Ba}$ ($t_{1/2} = 2.55$ min) daughter].

Appendix 6
CALIBRATION SOURCES FOR DETECTOR CALIBRATION

Nuclide	E_Y (keV)	$I_{Y,a}$ (%) ¹⁾	Nuclide	E_Y (keV)	$I_{Y,a}$ (%) ¹⁾
Am-241	59,5	35,3 ±0,5 (1,4)	Nb-95	765,8	99,80 ±0,04 (0,04)
Hg-203	72,9	12,8 ±0,2 (1,6)	Co-58	810,6	99,44 ±0,02 (0,02)
Co-57	122,1	85,54±0,25 (0,29)	Mn-54	834,8	99,978±0,002 (0,002)
Co-57	136,5	10,66±0,18 (1,7)	Co-56	846,8	99,974±0,001 (0,001)
Ni-56	158,3	98,7 ±0,1 (0,1)	Y-88	898,0	94,3 ±0,4 (0,4)
Se-75	264,7	59,1 ±0,2 (0,34)	Co-60	1173,2	99,86 ±0,02 (0,02)
Hg-203	279,2	81,5 ±0,2 (0,25)	Na-22	1274,5	99,95 ±0,03 (0,03)
I-131	364,5	82,5 ±0,4 (0,48)	Co-60	1332,5	99,986±0,001 (0,001)
Au-198	411,8	95,53±0,05 (0,05)	Na-24	1368,7	99,994±0,002 (0,002)
Na-22	511,0	181,08±0,40 (0,22)	La-140	1596,2	95,40 ±0,08 (0,08)
Sr-85	514,0	99,28±0,01 (0,01)	Y-88	1836,1	99,37 ±0,02 (0,02)
Sb-124	602,7	98,2 ±0,1 (0,1)	Na-24	2754,1	99,87 ±0,02 (0,02)
Cs-137	661,6	85,1 ±0,4 (0,5)			

¹⁾ The relative Error in % is in the ().

RECOMMENDED DECAY DATA

Nuclide: ^{152}Eu
 Half-life: $(13,52 \pm 0,02 \text{ years})$
 Photon Energy E and emission probabilities p:

Type of radiation	E (keV)	p (%)		E' (keV)	p'/p
Sm-K α	39.52/40.12	59.1	± 1.2		
Sm-K β	45.40/46.60	14.9	± 0.3		
γ	121.78	28.37	± 0.24	123.1	0.005^+
γ	244.69	7.51	± 0.06	247.9	0.003^+
γ	344.27	26.58	± 0.18		
γ	411.11	2.234	± 0.013	416.1	0.041
γ	443.98	3.121	± 0.018	444.0	
γ	778.89	12.96	± 0.07		
γ	964.12	14.62	± 0.06	963.4	
γ	1085.78	10.16	± 0.05	1089.8	0.168
γ	1112.02	13.56	± 0.06	1109.2	0.012
γ	1407.95	20.85	± 0.09		

This table contains only the 12 most intense lines out of 100 lines.

In column 4 of this table the energies E' of interfering energy lines are listed which may disturb the energy line E when calibrating Ge-detectors. Its relative emission probability p' with respect to the emission probability p of the main energy, is also given in the last column. At the energies 443.98 and 964.12 keV, p' is included in p. The + indicates that these lines originate from Eu-154, and the respective values p'/p assume a 0.3% fraction of the total Eu-154 activity.

The ranges given are for one standard deviation and exclude systematic errors.

Appendix 7

GAMMA RAY LINES OF FISSION PRODUCTS AND ACTINIDES

Nuclide	E_γ (keV)	$I_{\gamma,a} \pm \Delta I \left(\frac{\Delta I}{I} \cdot 100 \right)$ 1)
Kr-85	514.0	0.433 $\pm$ 0.010 (2.3)
Sr-91 2) + Y-91m 3)	555.6	65.5 $\pm$ 2.0 (3)
	652.3 4)	3.1 $\pm$ 0.3 (10)
	652.9	8.7 $\pm$ 0.9 (10)
	749.8	25.0 $\pm$ 1.3 (5)
	925.8	4.2 $\pm$ 0.4 (10)
	1024.3	36.0 $\pm$ 2.0 (5)
Y-91	1210.0	0.22 $\pm$ 0.02 (10)
Y-93 2)	266.8	7.15 $\pm$ 0.40 (5)
	947.1	1.96 $\pm$ 0.20 (10)
	1917.2	1.56 $\pm$ 0.16 (10)
Zr-95 + Nb-95m 3)	235.0	0.292 $\pm$ 0.015 (5)
	724.2	44.0 $\pm$ 0.5 (1.1)
	756.8	54.3 $\pm$ 0.5 (0.9)
Nb-95	765.8	99.8 $\pm$ 0.04 (0.04)
Zr-Nb-97 2)	507.6	5.15 $\pm$ 0.25 (5)
	657.9	105.84 $\pm$ 0.53 (0.5)
	743.3	92.5 $\pm$ 0.9 (1)
	1021.3 4)	1.25 $\pm$ 0.13 (10)
	1024.5 4)	1.20 $\pm$ 0.12 (10)
	1148.0	2.60 $\pm$ 0.3 (10)
Mo-99 + Tc-99m 3)	140.5	89.6 $\pm$ 0.5 (0.6)
	181.1	6.67 $\pm$ 0.17 (2.5)
	366.5	1.41 $\pm$ 0.06 (4.3)
	739.6	13.5 $\pm$ 0.5 (3.7)
	778.0	4.63 $\pm$ 0.20 (4.3)
Ru-103	479.1	89.35 $\pm$ 0.50 (0.6)
	610.4	5.72 $\pm$ 0.10 (1.8)
Rh-105	306.2	5.19 $\pm$ 0.16 (3.1)
	319.1	19.2 $\pm$ 0.5 (2.6)

Nuclide	E_γ (keV)	$I_{\gamma,a} \pm \Delta I \left(\frac{\Delta I}{I} \cdot 100 \right)$	1)
Ru-Rh-106	511.8	20.47 $\pm$ 0.23	(1.1)
	616.2	0.735 $\pm$ 0.012	(1.7)
	622.1	9.95 $\pm$ 0.08	(0.8)
	873.3	0.421 $\pm$ 0.006	(1.5)
	1050.0	1.452 $\pm$ 0.013	(0.9)
	1127.9	0.376 $\pm$ 0.006	(1.6)
	1561.9	0.148 $\pm$ 0.002	(1.1)
Sb-125	176.3	6.74 $\pm$ 0.10	(1.5)
	427.9	29.7 $\pm$ 0.3	(1.0)
	463.4	10.45 $\pm$ 0.13	(1.2)
	600.7	18.0 $\pm$ 0.6	(3.3)
	636.1	11.4 $\pm$ 0.3	(3)
Te-127m	417.9	0.97 $\pm$ 0.10	(10)
Te-129m	1084.0	0.61 $\pm$ 0.02	(3.3)
Te-131m	149.7	24.3 $\pm$ 1.5	(6)
	200.7	6.9 $\pm$ 0.7	(10)
	240.6	7.9 $\pm$ 0.8	(10)
	334.5	8.7 $\pm$ 0.7	(8)
	452.7	5.1 $\pm$ 0.5	(10)
	773.7	33.0 $\pm$ 3.0	(10)
	782.7	7.0 $\pm$ 0.7	(10)
	793.6	11.3 $\pm$ 1.1	(10)
	852.1	18.0 $\pm$ 1.0	(6)
	1125.5	10.4 $\pm$ 1.0	(10)
	1148.5	3.3 $\pm$ 0.3	(10)
	1206.5	8.2 $\pm$ 0.8	(10)
I-131	80.2	2.45 $\pm$ 0.09	(3.7)
	284.3	5.8 $\pm$ 0.2	(3.5)
	364.5	82.5 $\pm$ 0.4	(0.5)
	636.9	6.9 $\pm$ 0.2	(3)
	722.9	1.63 $\pm$ 0.05	(3)
Xe-131m	164.0	1.73 $\pm$ 0.20	(12)

Nuclide	E_γ (keV)	$I_{\gamma,a} \pm \Delta I \left(\frac{\Delta I}{I} \cdot 100 \right)$	1)
Te-I-132	228.2	85.0	± 5.0 (6)
	505.9	5.11	± 0.18 (3.5)
	522.7	16.5	± 0.6 (3.5)
	621.0	2.03	± 0.09 (4.5)
	630.2	13.8	± 0.5 (3.5)
	667.7 4)	101.3	± 0.1 (0.1)
	669.8 4)	5.4	± 1.4 (26)
	671.5 4)	5.3	± 0.6 (11)
	727.1	6.3	± 0.4 (6)
	772.6	78.0	± 1.7 (2.2)
	809.8 4)	3.09	± 0.29 (9)
	812.3 4)	6.0	± 0.4 (7)
	954.6	18.6	± 0.7 (4)
	1136.0	3.05	± 0.23 (7.5)
	1143.4	1.43	± 0.10 (7)
	1173.2	1.05	± 0.09 (9)
	1290.7	1.22	± 0.09 (7)
	1295.5	1.92	± 0.14 (7)
	1372.1	2.59	± 0.12 (4.6)
	1398.6	7.3	± 0.3 (4)
	1442.6	1.46	± 0.06 (4)
	1921.1	1.26	± 0.05 (4)
	2002.3	1.17	± 0.08 (7)
I-133 2)	529.9	87.5	± 0.9 (1)
	706.7	1.6	± 0.2 (10)
	856.4	1.25	± 0.12 (10)
	875.4	4.6	± 0.3 (7)
	1236.6	1.6	± 0.2 (10)
	1298.3	2.3	± 0.2 (10)
Xe-133	81.0	36.6	± 0.8 (2.2)
Cs-134	475.3	1.54	± 0.05 (3)
	563.2	8.56	± 0.24 (2.8)
	569.4	14.7	± 0.4 (2.7)
	604.6	97.54	± 0.10 (0.1)

Nuclide	E_γ (keV)	$I_{\gamma,a} \pm \Delta I \left(\frac{\Delta I}{I} \cdot 100 \right) \quad 1)$			
Cs-134 (continued)	795.8	85.4	± 0.9	(1.0)	
	802.1	8.73	± 0.24	(2.8)	
	1038.4	1.02	± 0.03	(3)	
	1167.7	1.93	± 0.06	(3)	
	1365.2	3.31	± 0.09	(3)	
I-135 2)	546.5	6.3	± 0.3	(5)	
	836.8	5.95	± 0.25	(4)	
	1038.7	7.45	± 0.25	(3.4)	
	1124.0	3.75	± 0.20	(5)	
	1131.6	21.35	± 0.42	(2)	
	1260.5	30.0	± 0.6	(2)	
	1457.7	8.8	± 0.4	(4)	
	1678.2	9.6	± 0.5	(5)	
	1706.7	4.14	± 0.20	(5)	
	1791.4	8.0	± 0.4	(5)	
Xe-135	249.9	91.4	± 0.5	(0.6)	
	608.1	2.6	± 0.5	(20)	
Cs-136	66.8	9.5	± 1.0	(10)	
	86.1	4.9	± 0.4	(7)	
	152.8	7.8	± 0.6	(8)	
	163.7	4.4	± 0.4	(9)	
	176.2	13.1	± 1.1	(8)	
	273.3	11.9	± 1.0	(8)	
	340.0	43.4	± 2.2	(5)	
	818.5	100.0	± 2.0	(2)	
	1047.7	79.4	± 2.3	(3)	
	1235.1	20.0	± 1.0	(5)	
Cs-137	661.6	85.1	± 0.4	(0.5)	
Ba-140	162.9	6.21	± 0.07	(1.1)	
	304.8	4.32	± 0.05	(1.2)	
	423.7	3.15	± 0.03	(1.0)	
	473.6	1.93	± 0.04	(2.0)	
	537.4	24.39	± 0.27	(1.1)	
La-140	242.0	0.473	± 0.028	(6)	
	266.6	0.452	± 0.025	(5.5)	
	328.8	20.81	± 0.21	(1.0)	

Nuclide	E_γ (keV)	$I_{\gamma,a} \pm \Delta I \left(\frac{\Delta I}{I} \cdot 100 \right)$	1)
La-140 (continued)	432.6	2.99 $\pm$ 0.04	(1.3)
	487.0	45.94 $\pm$ 0.46	(1.0)
	751.8	4.41 $\pm$ 0.04	(0.9)
	815.8	23.64 $\pm$ 0.17	(0.7)
	867.8	5.59 $\pm$ 0.05	(0.9)
	919.6	2.68 $\pm$ 0.03	(1.1)
	925.2	7.05 $\pm$ 0.09	(1.3)
	950.5	0.539 $\pm$ 0.019	(3.5)
	1596.6	95.40 $\pm$ 0.08	(0.08)
	2348.5	0.846 $\pm$ 0.015	(1.8)
	2522.0	3.433 $\pm$ 0.065	(1.9)
Ce-141	145.4	49.0 $\pm$ 1	(2)
Ce-143	231.6	1.98 $\pm$ 0.1	(5)
	293.3	41.3 $\pm$ 1.2	(3)
	350.6	3.2 $\pm$ 0.2	(6)
	490.4	1.8 $\pm$ 0.1	(5.5)
	664.6	5.9 $\pm$ 0.8	(14)
	722.0	5.6 $\pm$ 0.6	(10)
Ce-Pr-144	133.5	11.09 $\pm$ 0.16	(1.4)
	696.5	1.342 $\pm$ 0.013	(1.0)
	1489.2	0.279 $\pm$ 0.003	(1.1)
	2185.7	0.70 $\pm$ 0.01	(1.4)
Nd-147	91.1	26.8 $\pm$ 2.6	(10)
	531.0	11.7 $\pm$ 1.0	(9)
Pm-148m	288.2	12.8 $\pm$ 0.8	(6)
	414.1	19.7 $\pm$ 0.6	(3)
	432.8	5.3 $\pm$ 0.5	(10)
	501.3	6.95 $\pm$ 0.2	(3)
	550.2	94.3 $\pm$ 1.0	(1)
	599.6	11.5 $\pm$ 1.0	(9)
	611.3	5.3 $\pm$ 0.5	(10)
	629.9	87.4 $\pm$ 1.0	(1.1)
	725.6	30.8 $\pm$ 0.6	(2)
	915.0	18.8 $\pm$ 0.6	(3)
	1013.7	19.4 $\pm$ 0.6	(3)

Nuclide	E_γ (keV)	$I_{\gamma,a} \pm \Delta I \left(\frac{\Delta I}{I} \cdot 100 \right)$	1)
Pm-148	550.2	25.7	± 0.5 (2)
	611.3	1.23	± 0.1 (8)
	915.0	14.0	± 0.4 (3)
	1465.1	24.0	± 1.0 (4)
Pm-149 2)	285.9	3.1	± 0.3 (10)
Pm-151	340.1	21.0	± 0.5 (2.5)
Sm-153 2)	103.2	28.2	± 0.5 (2)
Eu-154	124.1	40.0	± 1.4 (3.5)
	247.9	6.67	± 0.22 (3.3)
	591.7	4.96	± 0.16 (3.2)
	692.4	1.70	± 0.05 (3)
	723.3	20.1	± 0.6 (3)
	756.8	4.34	± 0.13 (3)
	873.2	11.7	± 0.4 (3.4)
	996.3	10.2	± 0.3 (3)
	1004.8	17.5	± 0.5 (3)
	1274.5	34.0	± 1.0 (3)
	1490.2	0.64	± 0.02 (3.5)
	1596.5	1.71	± 0.06 (3.5)
Eu-155	86.5	32.6	± 1.0 (3)
Eu-156	89.0	8.5	± 0.4 (5)
	599.5	2.1	± 0.4 (20)
	646.3	6.5	± 0.3 (5)
	723.4	5.6	± 0.2 (3)
	811.8	9.7	± 0.3 (3)
	944.4	1.3	± 0.2 (15)
	960.5	1.5	± 0.2 (10)
	1065.1	4.9	± 0.2 (3)
	1079.2	4.2	± 0.3 (7)
	1153.4 4)	6.7	± 0.5 (7)
	1154.1 4)	4.9	± 0.3 (5)
	1230.7	8.3	± 0.3 (4)
	1242.4	6.7	± 0.4 (6)
	1277.4	3.0	± 0.2 (6)
	1366.4	1.7	± 0.1 (5)

Nuclide	E_γ (keV)	$I_{\gamma,a} \pm \Delta I \left(\frac{\Delta I}{I} \cdot 100 \right)$	1)
Eu-156 (continued)	1877.0	1.6 $\pm$ 0.2	(10)
	1966.0	3.9 $\pm$ 0.2	(5)
	2026.7	3.3 $\pm$ 0.1	(4)
	2097.7	3.9 $\pm$ 0.2	(5)
	2116.5	1.2 $\pm$ 0.4	(30)
	2180.9	2.3 $\pm$ 0.1	(5)
	2186.7	3.5 $\pm$ 0.3	(7)
Pa-233	86.7	2.05 $\pm$ 0.24	(12)
	271.6	0.34 $\pm$ 0.04	(12)
	300.1	6.5 $\pm$ 0.7	(10)
	311.9	38 $\pm$ 4	(10)
	340.5	4.33 $\pm$ 0.46	(11)
	375.4	0.65 $\pm$ 0.07	(11)
	398.5	1.37 $\pm$ 0.14	(10)
	415.8	1.71 $\pm$ 0.18	(11)
U-237	97.1	17.7 $\pm$ 1.8	(10)
	101.0	24.7 $\pm$ 2.5	(10)
	113.5	10.7 $\pm$ 1.1	(10)
	208.0	23.3 $\pm$ 0.5	(2.1)
Np-239	106.1	22.7 $\pm$ 2.5	(11)
	181.7	0.068 $\pm$ 0.018	(26)
	209.8	3.7 $\pm$ 0.5	(14)
	226.4 4)	0.19 $\pm$ 0.08	(40)
	228.2 4)	12.1 $\pm$ 1.4	(11)
	254.4	0.12 $\pm$ 0.06	(50)
	272.8	0.09 $\pm$ 0.05	(50)
	277.6	14.0 $\pm$ 1.4	(10)
	285.4	0.73 $\pm$ 0.12	(17)
	315.9	1.48 $\pm$ 0.21	(14)
	334.2	2.02 $\pm$ 0.25	(12)

1) $I_{\gamma,a}$ in % per decay

() Relative error in %

2) Estimated error

3) Daughter nuclide in equilibrium

4) Multiple peak in spectrum

Appendix 8

INFORMATION ON IMPORTANT FISSION PRODUCTS

1. Niobium-95

$$t_{1/2} = 34.97 \pm 0.03 \text{ days.}$$

Principal gamma ray line = 765.8 keV.

The majority of ^{95}Nb in irradiated fuels comes from the beta decay of ^{95}Zr . Zirconium-95 produced from the neutron capture of ^{94}Zr can also decay to ^{95}Nb , providing an additional source of ^{95}Nb . Niobium-95 activity becomes saturated relatively easily in the irradiation exposure of typical LWR fuel assemblies. Under certain conditions, ^{95}Nb migrates radially as well as axially in mixed oxide fuel pins. Often, the 765.8 keV peak of ^{95}Nb is summed with the 756.7 keV peak of ^{95}Zr as a monitor of the recent fission density of fuel pins. The only significant influence in unfolding spectra for ^{95}Nb is the 756.7 keV peak of ^{95}Zr , which should not cause any problems with present-day germanium detectors. The fission yield from ^{235}U is 6.5% and from ^{239}Pu is 4.9%.

2. Ruthenium-103

$$t_{1/2} = 39.35 \pm 0.05 \text{ days.}$$

Principal gamma ray lines = 497.1 and 610.2 keV.

The significant difference in fission yields of ^{103}Ru for ^{235}U (3.0%) and ^{239}Pu (6.8%) explains the increase of ^{103}Ru atomic density at higher burnup values. Radial⁴⁵ and axial migration of ^{103}Ru has been detected in experimental fuel pins. The 497-keV photopeak is easily resolved from interferences, whereas the 610.2-keV photopeak can suffer from interferences of ^{137}Cs (604.6 keV) and ^{106}Rh (616.3 keV). Because of the relatively short half-life of ^{103}Ru , it is seldom correlated with parameters for LWR fuel assemblies.

3. Zirconium-95

$$t_{1/2} = 63.98 \pm 0.06 \text{ days.}$$

Principal gamma ray lines = 724.2 and 756.7 keV

Zirconium-95 can be used as a monitor for locating the relative position of fuel material as well as the exact locations of interpellet gaps.

There has been no evidence that suggests significant radial or axial migration of the isotope. The fission yield for ^{235}U is 6.5% and for ^{239}Pu is 4.9%; the difference results in a decrease in the atomic density as burnup increases because of the increased contribution of ^{239}Pu fission to the total fissions. This difference in fission yields also explains the slope of ^{95}Nb as ^{95}Zr decays to ^{95}Nb . Both gamma ray peaks are easily resolvable.

4. Cerium-144

$$t_{1/2} = 284.5 \pm 1.0 \text{ day.}$$

Principal gamma ray lines = 696.4, 1489.2, and 2185.6 keV
from ^{144}Pr daughter.

Cerium-144 has been investigated as a potential burnup monitor because of its half-life, relative chemical inertness, and easily resolvable gamma ray spectra (^{144}Pr). For long irradiations, the half-life becomes a distinct disadvantage because the ^{144}Ce atomic density reflects the more recent exposure of the fuel assembly. Similar effects would be expected for the three previously discussed isotopes. Cerium-144 has been applied as a burnup monitor of CANDU fuels provided certain data are available for a generalized analysis of the reactor and fuel system.

5. Ruthenium-106

$$t_{1/2} = 366.5 \text{ days.}$$

Principal gamma ray lines = 622.2, 1050.5, and 1562.2 keV
from ^{106}Rh daughter.

The production of ^{106}Ru within irradiated fuel materials is very sensitive to the relative fission rates of ^{235}U and ^{239}Pu . The fission yield for ^{235}U is 0.40% and for ^{239}Pu is 4.3%; therefore, the atomic density of ^{106}Ru increases more rapidly than the corresponding increase in burnup. The influence of fission from ^{241}Pu can become significant at higher (~30,000 MWd/MTU) levels of burnup. The fission yield from ^{241}Pu is 6.2%, and with nearly 15% of the fission at 30,000 MWd/MTU being from ^{241}Pu . This effect cannot be ignored. This difference in fission yields can be used to estimate the relative number of fission events in ^{235}U and ^{239}Pu .

Significant radial and axial migration of ^{106}Ru has been detected. The 512 keV gamma ray peak of ^{106}Rh has possible interferences from ^{58}Co

and annihilation gammas of high-energy gamma ray lines (>1022 keV). The three lines listed are relatively easy to separate and give three signatures that can be compared to determine any anomalies.

6. Cesium-134

$$t_{1/2} = 2.062 \pm 0.005 \text{ years.}$$

Principal gamma ray lines = 604.7, 795.8, 801.8, 1038.5, 1167.9, and 1365.1 keV.

Cesium-134 is produced primarily by the neutron activation of ^{133}Cs that comes from ^{133}I through ^{133}Xe ; activation is primarily in resonance absorption. Iodine-133 has similar fission yields for ^{235}U (6.8%), ^{239}Pu (6.9%), and ^{241}Pu (6.7%). Therefore it is relatively independent of the fissioning source. Many examples of radial and axial migration have been detected. Migration appears to be a function of a variety of physical and chemical characteristics of the fuel material: thermal gradient, fuel density, and oxygen-to-metal ratio. Cesium-134 appears to migrate more readily than the ^{137}Cs isotope because of the longer half-lives of its precursors ^{133}I and ^{133}Xe . The large number of easily resolvable gamma ray peaks permit the use of ^{134}Cs to determine the intrinsic efficiency of the detector system under normal conditions.

7. Europium-154

$$t_{1/2} = 8.5 \pm 0.5 \text{ years.}$$

Principal gamma ray lines = 996.3, 1004.8, and 1274.4 keV.

The production of ^{154}Eu is very complicated. Europium-154 has essentially no direct production from the fissioning of ^{235}U and ^{239}Pu and results only from multiple neutron absorption of lower mass, high cross-section nuclides. The relative atomic density is non-linear over the range of burnup values 8000 to 32000 MWd/MTU. Europium has one of the longest half-lives of the measurable fission products, and it reflects the total irradiation history for typical LWR fuel assemblies.

The isotope does not appear to migrate significantly either radially or axially, but remains uniformly distributed under most irradiation conditions.

8. Cesium-137

$$t_{1/2} = 30.17 \pm 0.03 \text{ years.}$$

Principal gamma ray line = 661.6 keV (^{137m}Ba).

Cesium-137 has been investigated more than any other fission product because of its easily resolvable gamma ray and its long half-life. The fission yields for ^{235}U and ^{239}Pu are approximately the same, 6.3% and 6.7% respectively, with the yield of ^{241}Pu being 6.9%. The only distinct disadvantage of ^{137}Cs is its tendency to migrate radially and axially under certain conditions. Often, it is necessary to examine the entire fuel assembly to eliminate the effect of the axial migration. The production of ^{137}Cs in a typical BWR assembly indicates that a linear approximation should be acceptable over typical ranges of burnups and irradiation times.

Appendix 9
MASS YIELDS FOR FISSION PRODUCTS

	Th-232	U-233	U-235	U-238	Pu-239	Pu-241
A	Fast	Thermal	Thermal	Fast	Thermal	Thermal
85	3.93	2.19	1.33	0.75	0.566	0.392
91	7.26	6.52	5.93	4.40	2.53	1.82
92	7.49	6.65	5.98	4.65	3.05	2.25
93	7.21	7.04	6.39	5.00	3.92	2.93
94	6.23	6.81	6.45	5.00	4.48	3.37
95	5.30	6.21	6.54	5.13	5.07	3.99
97	3.96	5.39	6.00	5.45	5.70	4.75
99	2.76	4.89	6.16	6.10	6.33	6.19
102	0.5	2.42	4.19	6.17	6.09	6.32
103	0.146	1.60	3.12	6.36	6.90	6.60
105	0.05	0.54	0.90	3.81	5.47	6.67
106	0.041	0.255	0.387	2.85	4.6	6.08
125	0.033	0.110	0.028	0.076	0.110	0.042
127	0.089	0.50	0.11	0.12	0.55	0.18
129	0.36	1.56	0.64	0.9	1.65	0.80
130	0.8	2.4	1.5	1.7	2.6	1.65
131	1.60	3.54	2.93	3.20	3.90	3.13
132	2.83	4.84	4.35	5.16	5.43	4.64
133	3.96	6.03	6.73	6.56	7.11	6.72
134	5.32	6.15	7.68	7.68	7.73	8.09
135	5.30	6.27	6.55	6.77	7.65	7.11
136	5.38	6.82	6.25	6.84	7.14	7.29
137	6.45	6.85	6.26	6.00	6.67	6.58
140	8.31	6.45	6.36	5.96	5.62	5.92
141	7.28	6.60	5.85	5.50	5.29	4.88
143	7.12	5.88	5.92	4.60	4.48	4.51
144	7.66	4.64	5.44	4.64	3.76	4.17
147	2.97	1.80	2.22	2.54	2.09	2.24
148	2.18	1.30	1.67	2.10	1.68	1.90
149	1.44	0.76	1.05	1.80	1.24	1.46
150	1.09	0.52	0.644	1.28	1.01	1.18
151	0.41	0.32	0.407	0.81	0.76	0.90

	Th-232	U-233	U-235	U-238	Pu-239	Pu-241
A	Fast	Thermal	Thermal	Fast	Thermal	Thermal
152	0.32	0.22	0.262	0.53	0.58	0.72
153	0.21	0.107	0.163	0.42	0.44	0.53
154	0.06	0.045	0.072	0.22	0.273	0.38
155	0.01	0.026	0.032	0.12	0.17	0.23
156	0.0026	0.012	0.014	0.067	0.12	0.17

"Direct" fission yield factors used in the computer programme
(disregarding short-lived precursors)

Fission Product	Th-232 Fast	U-233 Thermal	U-235 Thermal	U-238 Fast	Pu-239 Thermal	Pu-241 Thermal
Kr-85m ⁴⁾	3.93	2.19	1.33	0.75	0.566	0.392
Kr-85 ⁴⁾	0.83	0.507	0.288	0.16	0.129	0.086
Sr-91	7.26	6.50	5.93	4.40	2.52	1.82
Y-91	0	0.02	0	0	0.01	0
Te-131m ⁴⁾	0.12	0.75	0.39	0.32	0.64	0.34
I-131 ⁴⁾	1.40	2.79	2.54	2.88	3.26	2.79
Te-132	2.70	4.57	4.33	5.17	5.28	4.61
Xe-132 ¹⁾	0	0.27	0.02	0	0.15	0.03
I-135	4.62	4.95	6.32	6.74	6.50	6.82
Xe-135 ²⁾	0.03	1.32	0.23	0.03	1.15	0.29
Ba-140	8.31	6.35	6.36	6.00	5.58	5.92
La-140	0	0.10	0	0	0.04	0
Pm-148m	0	0.0034	0	0	0.0016	0
Pm-148	0	0.0003	0	0	0.0002	0
Nd-150	1.09	0.48	0.643	1.28	0.97	1.17
Sm-150 ³⁾	0	0.04	0.001	0	0.04	0.01

¹⁾ Sum of direct fission yields of I-132 and Xe-132

²⁾ Sum of direct fission yields of Xe-135m and Xe-135

³⁾ Sum of direct fission yields of Pm-150 and Sm-150

⁴⁾ see text

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